Sequences and Series

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Chapter 1

Foundations

1.1 Sets

A **set** is a collection of objects. These objects are called **elements** or **members** of the set. A set with no objects is called the **empty set** and is denoted by $\emptyset$.

Often when we are dealing with *finite* sets (sets that have finitely many elements) we simply describe them by listing their elements inside "curly brackets". For example, the set whose elements are the numbers 1, 6 and 2 is written

$$\{1, 6, 2\}.$$

For a set A we write $x \in A$ if x is an element of A. By $x \notin A$ we mean that x is not an element of A.

Definition 1 *If A and B are sets with the property that every element of A belongs to B; i.e.*

$$x \in A \Rightarrow x \in B,$$

then we say that A is a subset *of B (or A is contained in B). If A is a subset of B we often write $A \subseteq B$. The empty set $\emptyset$ is a subset of any nonempty set.*

If $A \subseteq B$ and $A \neq B$, then A is said to be a proper subset of B, denoted as $A \subset B$.

By $B \supseteq A$ we mean $A \subseteq B$; by $B \supset A$ we mean $A \subset B$. We say $A = B$ if $A \subseteq B$ and $B \subseteq A$. Thus,

$$\{1, 6, 2\} = \{6, 1, 2\}.$$

The empty set $\emptyset$ is a subset of any given set.

Given a set Ω, we can define a subset of Ω in terms of a property say P:

$$\{x \in \Omega : x \text{ has property } P\}.$$

Definition 2 *Consider two subsets A, B of a set Ω.*

1. *A* **union** *of two sets A and B is defined as*

$$A \cup B := \{x \in \Omega : x \in A \text{ or } x \in B\}.$$

2. *An* **intersection** *of two sets A and B is defined as*

$$A \cap B := \{x\Omega : x \in A \text{ and } x \in B\}.$$

3. *A* complement of B relative to A *(or* **set-theoretic difference** *of A and B) is defined as*

$$A \setminus B := \{x \in A : x \in A \text{ and } x \notin B\}.$$

4. *The set $\Omega \setminus B =: B^c$ is called the* **complement** *of B.*

5. *We say sets A and B are* **disjoint** *if $A \cap B = \emptyset$.*

Notation:

- $\mathbb{N} := \{1, 2, 3, \ldots\}$ is the set of natural numbers.
- $\mathbb{Z}$ denotes the set of integers.
- $\mathbb{Q}$ denotes the set of rational numbers.

Notice that

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q}.$$

Remark 1 *We assume that all the familiar arithmetics involving rational numbers are known, and will use (and in fact have used) them freely: e.g., $\frac{m}{n} + \frac{p}{q} = \frac{mq+np}{nq}$ for integers m, n, p, q with $n, q \neq 0$. We also assume that $p < q, p > q, p \leq q, p \geq q$ are understood for $p, q \in \mathbb{Q}$. Loosely speaing, we assume anything about rational numbers are known.*

1.2 Ordered set

Definition 3 *Order Let S be a set. A (strict total) order on S is a relation, denoted by $\prec$, with the following properties:*

- *For $x, y \in S$, one and only one of the following holds:*

$$x \prec y \text{ (“}x \text{ precedes } y\text{”)}, x = y, \ y \prec x \text{ (“}y \text{ precedes } x\text{”)}.$$

- *For $x, y, z \in S$, if $x \prec y$ and $y \prec z$, then $x \prec z$.*

Here the order $\prec$ is just a relation, does not necessarily mean “strictly smaller than”.
Examples of $\prec$:

Example 1 *Consider the set of rational numbers $\mathbb{Q}$, and $>$ means “(strictly) bigger than”. Then $>$ is an order on $\mathbb{Q}$. Indeed, for any $x, y \in \mathbb{Q}$, either $x > y$, or $x = y$ or $y > x$ holds, and only one of them holds; moreover, if $x > y$ and $y > z$ then $x > z$.*

Example 2 *Consider the set of rational numbers $\mathbb{Q}$, and $<$ means “smaller than”. Then $<$ is an order on $\mathbb{Q}$. Indeed, for any $x, y \in \mathbb{Q}$, either $x < y$, or $x = y$ or $y < x$ holds, and only one of them holds; moreover, if $x < y$ and $y < z$ then $x < z$.*

Examples of relations that are not orders:

Example 3 *Consider the set of rational numbers $\mathbb{Q}$. Then $\leq$ is not an order on $\mathbb{Q}$. Indeed, for any $x \in \mathbb{Q}$, tale $y = x$, all of then three assertions hold: $x \leq y$, $x = y$, $y \leq x$ holds, but the definition requires exactly one of the three holds.*

Example 4 *Consider the set of rational numbers $\mathbb{Q}$. Then $\neq$ is not an order on $\mathbb{Q}$. Indeed, if $x \neq y$ and $y \neq z$, it does not necessarily hold that $x \neq z$. For example, take $x = z = 1$ and $y = 0$.*

Example 5 *Let $S = \mathcal{P}(\mathbb{Q})$, which means the set of all subsets of $\mathbb{Q}$. Then $\subset$ is not an order on S. Indeed, consider $A = \{1, 2\}$ and $B = \{2, 3\}$. Then $A \subset B$ does not hold; $B \subset A$ does not hold; and $A = B$ does not hold. So the first requirement in the definition of an order is not satisfied by $\subset$, despite the second requirement in the definition of an order holds for $\subset$.)*

Definition 4 (Ordered set) *A set S endowed with an order $\prec$ is called an ordered set, denoted as $(S, \prec)$. When the order is clear, simply call S an ordered set.*

Suppose $(S, \prec)$ or for brevity S is an order set in this section.

- It is convenient to introduce notations:
  - $\preceq$: $x \preceq y$ if and only if $x \prec y$ or $x = y$.

- $\succ$; $y \succ x$ (y succeeds (is after) x) if and only if $x \prec y$.
- $\succeq$: $y \succeq x$ if and only if $y \succ x$ or $y = x$.

Definition 5 (Upper bound) *Consider an ordered set S and $\emptyset \neq E \subseteq S$. If there is $y \in S$ such that for all $x \in E$, it holds that $x \preceq y$, then E is said to be bounded above, and y is called an upper bound of E.*

Example 6 *Consider the ordered set $(\mathbb{Q}, <)$. Then $\{0, 1, 2\}$ is bounded from above, with an upper bound e.g.,* 3.

Definition 6 (Least upper bound) *Suppose S is an ordered set, and $\emptyset \neq E \subseteq S$ is bounded above. Suppose there exists some $\alpha \in S$ such that*

- *α is an upper bound of E.*
- *For any $\gamma \prec \alpha$, γ is not an upper bounded of E.*

Then α is called the least upper bound of E (in S), or the supremum of E, denoted as

$$\alpha = \sup E.$$

In the above definition, one may ask if $\sup E$ is unique, if it exists. The answer is affirmative. Indeed, if β is also a least upper bound of E, then the definition of an order implies one and only one of the following holds:

- $\alpha \prec \beta$, in which case, α is not an upper bound (by definition of least upper bound), which cannot be true;
- $\beta \prec \alpha$, in which case, β is not an upper bound (by definition of least upper bound), which cannot be true;
- $\beta = \alpha$.

Moreover, if $\sup E$ exists, then it is given by the least element of F, the set of all upper bounds of E in S. Indeed, $\sup E \in F$ by definition of the least upper bound, and if there is some $\alpha' \in F$ such that $\alpha' \prec \sup E$, then by definition of the least upper bound, α' is not an upper bound of E, so that α' is not in F, which is a contradiction. Thus, F does not have any element preceding $\sup E$. Since $(S, \prec)$ is an ordered set, any element F must be succeeding $\sup E$, and $\sup E$ is the least element of F.

Consider the ordered set $(\mathbb{Q}, <)$. Then the least upper bound of $E \subset \mathbb{Q}$, if exists, is just the smallest element of the set of all upper bounds of E in $\mathbb{Q}$.

Example 7 *Consider the ordered set $(\mathbb{Q}, <)$. The set $\{0, 1, 2\} \subset \mathbb{Q}$ has the least upper bound* 2. *Indeed, the set of upper bounds of $\{0, 1, 2\}$ in $\mathbb{Q}$ is $\{x \in \mathbb{Q} : \ x \geq 2\}$, and the smallest element of this set is given by* 2.

The next example shows that not all bounded above subsets of $\mathbb{Q}$ have a least upper bound in $\mathbb{Q}$.

We need a preliminary observation:

Lemma 1 *There is no rational number $x \in \mathbb{Q}$ satisfying $x^2 = 2$.*

Proof. Suppose $x \in \mathbb{Q}$ such that $x^2 = 2$. Write $x = \frac{m}{n}$ with m and n being coprime[1]. So $(\frac{m}{n})^2 = 2$ or $m^2 = 2n^2$. Hence m^2 is divisible by 2 (i.e. m^2 is an even number) and so m is divisible by 2 (for if m is odd, then m^2 is odd too). Write $m = 2k$ and so $(2k)^2 = 2n^2$. Divide by 2 and note that $2k^2 = n^2$, and hence n is divisible by 2. But that is a contradiction against m, n being coprime. □

Example 8 *Consider the ordered set $(\mathbb{Q}, <)$, and let $E = \{p \in \mathbb{Q} : \ p > 0, p^2 < 2\}$. Then $E \neq \emptyset$ as $1 \in E$. Moreover, E is bounded above by e.g.,* 4. *Indeed, if there is some $p \in E$ such that $4 < p$, then $16 = 4^2 < p^2 < 4$, which yields a contradiction. The collection of the upper bounds of E in $\mathbb{Q}$ is given by $F = \{p \in \mathbb{Q} : \ p > 0, p^2 > 2\}$, because no positive rational number p satisfies $p^2 = 2$. See a more detailed justification below. We will show that F does not have a smallest element. That means, for any $p \in F$, we can find some $q \in F$ such that $q < p$. This would in turn imply that E does not have a least upper bound in $\mathbb{Q}$.*

[1] In number theory, two integers a and b are said to be coprime (also written co-prime) if the only positive integer (factor) that divides both of them is 1.

Proof. First, we verify that the collection of upper bounds of E is indeed given by F. Each $p \in F$ is an upper bound of E, because, otherwise, there would be some $q \in E$ such that $p < q$, and then $2 < p^2 < q^2 < 2$, a contradiction. On the other hand, any upper bound of E in $\mathbb{Q}$ must be in F. This can be seen as follows.

Suppose for contradiction that there is such a rational number $y > 0$. Then y can only be in E, because no rational number p satisfies $p^2 = 2$. Consider $y + \frac{1}{n}$ for some positive integer n. We would like to show that if this n is large enough, then $y + \frac{1}{n}$ is still in E. This would contradiet that y is an upper bound of E.

Such an n can be found as follows:

$$(y + \frac{1}{n})^2 < 2 \Leftrightarrow y^2 + \frac{1}{n^2} + \frac{2y}{n} < 2 \Leftrightarrow \frac{1}{n^2} + \frac{2y}{n} < 2 - y^2.$$

Note that $\frac{1}{n^2} + \frac{2y}{n} < \frac{1+2y}{n}$. So if we take n such that $\frac{1+2y}{n} < 2-y^2$, i.e., $n > \max\{1, \frac{1+2y}{2-y^2}\}$, then $\frac{1}{n^2} + \frac{2y}{n} < 2-y^2$ and thus $(y + \frac{1}{n})^2 < 2$. For such a positive integer n, $y + \frac{1}{n}$ is in E, as required.

Second, we show that F has no minimal element.

Let $p \in F$ be fixed. Then we construct the required $q \in F$ satisfying $q < p$. That means, we need a positive rational number $q < p$ satisfying $q^2 > 2$.

Since $p^2 > 2$, we have $p^2 - 2 > 0$. We would like to find some positive $x \in \mathbb{Q}$ such that $q = p - x$, that is,

- $p - x > 0$;
- $(p - x(p^2 - 2))^2 > 2$.

Take $x = \frac{p^2-2}{p+2}$. Then

$$p - x = p - \frac{p^2 - 2}{p + 2} = \frac{p^2 + 2p - p^2 + 2}{p + 2} = \frac{2p + 2}{p + 2} > 0;$$

and

$$(p - x)^2 = (p - \frac{p^2 - 2}{p + 2})^2 = \left(\frac{p^2 + 2p - p^2 + 2}{p + 2}\right)^2 = \left(\frac{2(p + 1)}{p + 2}\right)^2$$

and

$$\begin{aligned} &\left(\frac{2(p + 1)}{p + 2}\right)^2 > 2 \Leftrightarrow 4(p^2 + 1 + 2p) > 2(p^2 + 4 + 4p) \\ \Leftrightarrow \quad &2p^2 > 4 \Leftrightarrow p^2 > 2, \end{aligned}$$

which is true. □

Definition 7 (Least upper bound property) *An ordered set S is said to have the least upper bound property if any nonempty bounded above $E \subseteq S$ has a least upper bound in S.*

The previous example demonstrates that $(\mathbb{Q}, <)$ does not have the least upper bound property.

We can similarly define lower bound, and greatest lower bound.

Definition 8 (Lower bound) *Consider an ordered set S and $\emptyset \neq E \subseteq S$. If there is $y \in S$ such that for all $x \in E$, it holds that $x \succeq y$, then E is said to be bounded below, and y is called a lower bound of E.*

Definition 9 (Greatest lower bound) *Suppose S is an ordered set, and $\emptyset \neq E \subseteq S$ is bounded below. Suppose there exists some $\alpha \in S$ such that*

- *α is a lower bound of E.*
- *For any $\gamma \succ \alpha$, γ is not a lower bounded of E.*

Then α is called the greatest lower bound of E, or the infimum of E, denoted as

$$\alpha = \inf E.$$

If the ordered set has the least upper bound property, then it also has the greastest lower bound property in the sense of the following theorem.

Theorem 1 (Greatest lower bound property) *Suppose $(S, \prec)$ is an ordered set and has the least upper bound property. Then for any nonempty $B \subseteq S$, which is bounded below, it has the greatest lower bound in S.*

Moreover, if L is the set of lower bounds of B in S, then $\sup L$ *exists, and*

$$\sup L = \inf B.$$

Proof. L is bounded above by any given element of B. Therefore, we may apply the least upper bound property and see $\sup L$ exists. Now we show that $\sup L = \inf B$.

First, we show that $\sup L$ is a lower bound of B. Suppose this is not the case for contradiction. Then there is some $b \in B$ such that $\sup L \succ b$, i.e., $b \prec \sup L$. Then from the definition of $\sup L$, we see that b cannot be an upper bound of L. So there is some $l \in L$ such that $b \prec l$. However, this contradicts that $l \in L$ is a lower bound of B. Consequently, $\sup L$ is a lower bound of B.

Second, we show for each $x \succ \sup L$, x is not a lower bound of B. Let such an $x \succ \sup L$ be fixed. We have that $x \notin L$, because $\sup L$ is an upper bound of L. Then x is not a lower bound of B by the definition of L. □

1.3 Field

Definition 10 (Field) *A field is a set F endowed with two operations: one is called addition $+$ and one is called multiplication $\cdot$ (also denoted as $\times$), which satisfy the following axioms:*

- *Axioms for addition:*
  - *For any $x, y \in F$, $x + y \in F$.*
  - *Commutative: $x + y = y + x$.*
  - *Associative: $(x + y) + z = x + (y + z)$ for all $x, y, z \in F$.*
  - *F contains an element, called 0, such that $0 + x = x$ for all $x \in F$.*
  - *For each $x \in F$, there is an element $-x \in F$ such that $x + (-x) = 0$.*
- *Axioms for multiplication:*
  - *For any $x, y \in F$, $x \cdot y \in F$. (When there is no confusion, we write $x \cdot y$ as xy).*
  - *Commutative:$x \cdot y = y \cdot x$ for all $x, y \in F$.*
  - *Associative: $(x \cdot y) \cdot z = x \cdot (y \cdot z)$ for all $x, y, z \in F$.*
  - *F contains an element, called $1 \neq 0$, such that $1 \cdot x = x$ for all $x \in F$.*
  - *If $x \in F$ and $x \neq 0$, then there is an element $\frac{1}{x} \in F$ such that $x \cdot \frac{1}{x} = 1$.*
- *Distributive law: $x \cdot (y + z) = x \cdot y + x \cdot z$ for all $x, y, z \in F$.*

It is conventional to use the following notations:

- $x + (-y) =: x - y$.
- $x \cdot \frac{1}{y} =: \frac{x}{y} =: x/y$, if $y \neq 0$.
- $(x + y) + z =: x + y + z$.
- $(x \cdot y) \cdot z =: x \cdot y \cdot z$.
- $x \cdot x =: x^2$, $x \cdot x \cdot x =: x^3$,....
- $x + x =: 2x$, $x + x + x =: 3x$, ...

Example 9 $\mathbb{Q}$ *with the usual addition and multiplication is a field, where* 0, 1, $\frac{1}{x}$ *are understood in the usual sense. The set of natural numbers* $\mathbb{N}$ *with the usual addition and multiplication is, clearly, not a field.*

Remark 2 *It is possible that a field F has finitely many elements.*

Example 10 *Consider $F = a, b$. Define $+$ and $\cdot$ on F as follows:*

$$a + a := a, a + b := b, b + a := b, b + b := a;$$
$$a \cdot a := a, a \cdot b := a, b \cdot a := a, b \cdot b := b.$$

Then $(F, +, \cdot)$ is a field. The 0 element is a, and associated with b, the $-b$ element is b.

Consider a field $(F, +, \cdot)$. A subset $E \subseteq F$ is called a subfield of F if $(E, +, \cdot)$ is a field.

Theorem 2 (Consequences of axioms of addition) *Axioms of addition imply: for $x, y, z \in F$ where F is a field,*

(a) If $x + y = x + z$, then $y = z$.

(b) If $x + y = x$, then $y = 0$. (So 0 is unique in F.)

(c) If $x + y = 0$, then $y = -x$. (So $-x$ is unique in F corresponding to x.)

(d) $-(-x) = x$.

Proof. (a) Add $(-x)$, which exists due to the axioms of addition, to te both sides of $x + y = x + z$ and use the associative law: the left hand side becomes $(-x) + (x + y) = ((-x) + x) + y = 0 + y = y$; and the right hand side becomes $(-x) + (x + z) = ((-x) + x) + z = 0 + z = z$. Thus, $y = z$.
(b) Follows from (a) by putting $z = 0$.
(c) Follows from (a) by taking $z = -x$.
(d) We know $x + (-x) = 0$. (c) implies $x = -(-x)$. □

Theorem 3 (Consequences of axioms of multiplication) *Axioms of multiplication imply: for $x, y, z \in F$ where F is a field,*

(a) If $x \neq 0$, $x \cdot y = x \cdot z$, then $y = z$.

(b) If $x \neq 0$, $x \cdot y = x$, then $y = 1$. (So 1 is unique in F.)

(c) If $x \neq 0$, $x \cdot y = 1$, then $y = \frac{1}{x}$. (So $\frac{1}{x}$ is unique in F corresponding to $x \neq 0$.)

(d) If $x \neq 0$, then $\frac{1}{1/x} = x$.

The proof of this statement is left as an exercise.

Remark 3 *When there is no danger of confusion, we write $x \cdot y$ as xy in what follows.*

Theorem 4 *Consequences of axioms of field Axioms of field imply: for $x, y, z \in F$ where F is a field,*

(a) $0x = 0$. In particular, if $x \neq 0$, then $\frac{1}{x} \neq 0$.

(b) If $x, y \neq 0$, then $xy \neq 0$.

(c) $(-x)y = -(xy) = x(-y)$.

(d) $(-x)(-y) = xy$.

Proof (a) We know if $x + y = x$ then $y = 0$ (this is part (b) of consequences of axioms of addition). We use this to prove (a) as follows. Since by distributive law, $0x + 0x = (0 + 0)x = 0x$, we see $0x = 0$.
(b) Suppose for contradiction that $xy = 0$. Then $(\frac{1}{x})xy\frac{1}{y} = \frac{1}{x}0\frac{1}{y}$. But the left hand side is 1 and the right hand side is 0 (by part (a)). This is a desired contradiction (as $1 \neq 0$), meaning $xy \neq 0$.
(c) Prove $(-x)y = -(xy)$ as follows. We need to show $(-x)y + (xy) = 0$. The left hand side is, by distributive law $(-x + x)y = 0y = 0$, the last equality valid by (a). This implies $-(xy) = (-x)y$.
Prove $x(-y) = -(xy)$ similarly: $x(-y) + (xy) = x(-y + y) = x0 = 0x = 0$.
(d) By part (c) $(-x)(-y) = -(x(-y)) = -(-(xy)) = (xy)$, where the last equalty holds by part (d) of consequences of axioms of addition. □

Definition 11 (Ordered field) *An ordered field F is a field, on which an order $\prec$ is endowed, and satisfies*

- *If $y \prec z$, then $x + y \prec x + z$ for all $x, y, z \in F$.*
- *If $x, y \in F$, $0 \prec x$ and $0 \prec y$, then $0 \prec x \cdot y$.*

In an ordered field, if $0 \prec x$, then we call $x \in F$ positive; and if $x \prec 0$, then we call $x \in F$ negative.

The following statement is a consequence of the defintion of ordered field.

Theorem 5 (Properties of ordered field) *Let F be an ordered filed with order $\prec$ and $x, y \in F$. Then the following hold.*

(a) If $x \succ 0$, then $-x \prec 0$ and vice versa.

(b) If $x \succ 0$ and $y \prec z$, then $xy \prec xz$.

(c) If $x \prec 0$ and $y \prec z$, then $xy \succ xz$.

(d) If $x \neq 0$, then $x^2 \succ 0$. (In particular, $1 \succ 0$.)

(e) If $0 \prec x \prec y$, then $0 \prec \frac{1}{y} \prec \frac{1}{x}$.

Proof. (a) By the first additional requirement in the definition of ordered field, $0 \prec x$ implies $0+(-x) \prec x+(-x)$, i.e., $-x \prec 0$.

If $x \prec 0$, then similarly, $(-x) + x \prec (-x) + 0$ and so $0 \prec -x$.

(b) Since $y \prec z$, $(z - y) \succ 0$, and so by the second additional requirement in the definition of ordered field, $x(z-y) \succ 0$. That is, $xz - xy \succ 0$. By the first extra requirement in the definition of ordered field, this implies $xz - (xy) + (xy) \succ 0 + xy = xy$, and thus $xz \succ xy$.

(c) $x \prec 0$ implies $-x \succ 0$ (by part (a)). Part (b) implies $(-x)y \prec (-x)z$. Thus, $-(xy) \prec -(xz)$. Now the first additional requirement in definition of ordered field implies $xy - (xy) \prec (xy) - (xz)$, i.e., $0 \prec (xy) - (xz)$. Again using the first additional requirement in definition of ordered filed: $0 + (xz) \prec (xy) - (xz) + (xz)$, i.e., $xz \prec xy$ as required.

(d) If $x \succ 0$, then by part (b), $x^2 = x \cdot x \succ x0 = 0$.

If $x \prec 0$, then $-x \succ 0$. Part (d) of consequences of axioms of field implies $(-x)(-x) = x \cdot x = x^2$. Now the previous reasoning applied to $(-x) \succ 0$ yields $(-x)(-x) \succ 0$, and so $x^2 \succ 0$.

(e) Firstly, we show $\frac{1}{y} \succ 0$. Since $0 \prec y$, part (c) asserts that if $\frac{1}{y} \prec 0$, then $1 = y \cdot \frac{1}{y} \prec 0$, which is a contradiction because $1 \succ 0$. Therefore, $\frac{1}{y} \succ 0$.

Next show $\frac{1}{y} \prec \frac{1}{x}$ as follows. Since $0 \prec x$, we see $0 \prec \frac{1}{x}$. Now by part (b), from $x \prec y$, we get

$$\frac{1}{x}\frac{1}{y}x \prec \frac{1}{x}\frac{1}{y}y.$$

Using the commutative and associative law of multiplication, we get $\frac{1}{y} \prec \frac{1}{x}$ as required. □

1.4 Real field and real numbers

Now we can define the set of real numbers, $\mathbb{R}$, as an ordered field with the least upper bound property.

Definition 12 (Real field) *An ordered (with the order $<$) field with the least upper bound property and contains $\mathbb{Q}$ (endowed with the usual addition and multiplication) as a subfield is called the real field, denoted as $\mathbb{R}$ or $(-\infty, \infty)$. Here by subfield we mean that applying the addition and multiplication to the elements of $\mathbb{Q}$ is the same as applying the usual addition and multiplication to elements of $\mathbb{Q}$.*

Any element of $\mathbb{R}$ is called a real number. Of course, $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$. The last inclusion is strict because, as was demonstrated earlier, $\mathbb{Q}$ does not have the least upper bound property.

While we do not explore any further details, we make the following remark.

- The real field really exists and is unique in a certain sense. One way of constructing the real field starting from $\mathbb{Q}$ is the method of Dedekind cut.

Remark 4 *Below, unless specified otherwise, when we talk about least upper bound or greatest lower bound of a subset of $\mathbb{R}$, we always mean them in $\mathbb{R}$.*

A useful property of $\mathbb{R}$ is the following one.

Theorem 6 (Archimedean property) *Suppose $x, y \in \mathbb{R}$ and $x > 0$. Then there is some positive integer n such that*

$$nx > y.$$

The above holds trivially if $y \leq 0$. Suppose $y > 0$. Then the interpretation of the above result is that given a segment of length y and another segment of length x. Then one can form a new segment by joining n copies of the segment of length x, such that the length of the resulting segment is bigger than y.

The next statement further reinforces the relationship between $\mathbb{R}$ and $\mathbb{Q}$.

Theorem 7 (Denseness of $\mathbb{Q}$ in $\mathbb{R}$) *Suppose $x, y \in \mathbb{R}$, and $x < y$. Then there is some rational number $p \in \mathbb{Q}$ such that*

$$x < p < y.$$

Example 11 (Supremum or infimum for E (even if they exist) need not be in E)

- *For example, the set $E := \{x \in \mathbb{Q} : x < 1\}$ has a least upper bound of 1. Indeed, 1 is clearly an upper bound of E. Moreover, if $y \in \mathbb{R}$ is such that $y < 1$, then by the denseness of $\mathbb{Q}$ in $\mathbb{R}$, there would be some $z \in \mathbb{Q}$ such that $y < z < 1$. This rational number z is in E, meaning that y cannot be an upper bound of E. Therefore, 1 is the least upper bound of E (in $\mathbb{R}$).*

  Note that 1 is not in the set E.

- *On the other hand, if we take $G := \{x \in \mathbb{Q} : x \leq 1\}$, then the least upper bound of G is clearly also 1, and in this case $1 \in G$.*

- *The set $P := \{x \in \mathbb{Q} : x \geq 0\}$ has no upper bound. Indeed, for any positive real number $y > 0$, the Archimedean property of $\mathbb{R}$ implies that for some positive integer n, $y < n \cdot 1 = n$. But $n \in P$. This means, y cannot be an upper bound of P. Therefore, P cannot have a least upper bound. On the other hand 0 is the greatest lower bound of P.*

As a consequence of the denseness of $\mathbb{Q}$ in $\mathbb{R}$, we have

Corollary 1 *Suppose x, y are two real numbers satisfying: for each rational number $\epsilon > 0$, if there are two real numbers a_ϵ and b_ϵ such that $b_\epsilon - a_\epsilon < \epsilon$ and $a_\epsilon \leq x, y \leq b_\epsilon$, then $x = y$.*

Proof. Let us prove $x = y$ given the conditions in the statement. Suppose $x \neq y$ for contradiction. Then $x > y$ or $x < y$. Consider the case of $x > y$. Then $x - y > 0$. The denseness of $\mathbb{Q}$ implies that there is some rational number z such that $0 < z < x - y$. Let $\epsilon = z$, and take the corresponding a_ϵ, b_ϵ. Note that $x - y \leq b_\epsilon - a_\epsilon \leq z$, but $z < x - y$, yielding a contradiction. The case of $x < y$ can be handled in the same way by swapping the roles of x and y. □

The following form of the above corollary is often used: Suppose x, y are two real numbers satisfying: for each natural number $n > 0$, if there are two real numbers a_n and b_n such that $b_n - a_n < \frac{1}{n}$ and $a_n \leq x, y \leq b_n$, then $x = y$. Indeed, for any rational number $\epsilon > 0$, we know, by taking $n > 0$ as a natural number satisfying $n > \frac{1}{\epsilon}$ (such an n exists e.g., by the Archimedean property), we have $\epsilon < \frac{1}{n}$. Then apply the previous corollary.

In particular, if $0 \leq x < \frac{1}{n}$ for all natural number $n > 0$, then $x = 0$.

We have seen that there is no positive natural number $x \in \mathbb{Q}$ satisfying $x^2 = 2$. The next statement asserts that this equation has a unique positive real solution.

Theorem 8 (nth root of a real number) *For each positive real number $x > 0$, and positive integer n, there is one and only one positive real number $y \in \mathbb{R}$ satisfying $y^n = x$.*

In this case, we denote $y := x^{\frac{1}{n}}$ or $y := \sqrt[n]{x}$. When $n = 2$, $\sqrt[2]{x}$ is written as $\sqrt{x}$.

Proof. Let $A = \{p \in \mathbb{Q} : \ p > 0, p^n \leq x\} \cup \{y \in \mathbb{Q} : y \leq 0\}$. Let $\bar{A} = \{q \in \mathbb{Q} : \ q > 0, \ q^n > x\}$. A and $\bar{A}$ are not empty, because there exist a natural number $m > 0$ such that $\frac{1}{m} < x < m$ by the denseness of $\mathbb{Q}$ and by the Archimedean property, and then $\left(\frac{1}{m}\right)^n \leq \frac{1}{m} < x < m \leq m^n$ with $\frac{1}{m}$ and m both being rational numbers. It also follows that A is bounded above (by such an m). Therefore, by the least upper bound property of $\mathbb{R}$, we may take $\xi = \sup A > 0$.

Let us show that $\xi^n = x$ as follows.

Let $\epsilon > 0$ be fixed. By the Archimedean property, $\xi < 2N$ for some natural number $N > 0$. Let $\delta > 0$ be such that

- $\xi + \delta < 2N$.
- $n(2N)^{n-1}\delta < \epsilon$.
- $\xi > \frac{\delta}{2}$.

(Just take $0 < \delta < \min\{1, \frac{\epsilon}{n(2N)^{n-1}}, 2\xi\}$.)

By the denseness of $\mathbb{Q}$ in $\mathbb{R}$, there are rational numbers p and q such that

$$0 < \xi - \frac{\delta}{2} < p < \xi < q < \xi + \frac{\delta}{2}.$$

Then $p^n < \xi^n < q^n$.

Note that $p \in A$, because $p < \xi = \sup A$, meaning that $p < p'$ for some $p' \in A$, and so $p^n < (p')^n \le x$ and thus $p \in A$. Also, $q \in \bar{A}$, for otherwise, q can only be in A, and this contradicts $\xi = \sup A$ because $q > \xi$.

Now, from $q \ge p$ we see

$$\begin{aligned} & q^n - p^n = (q-p)(q^{n-1} + q^{n-2}p + q^{n-3}p^2 + \cdots + qp^{n-2} + p^{n-1}) \le (q-p)(nq^{n-1}) \\ \le \quad & \delta(n(\frac{\delta}{2} + \xi)^{n-1}) \le \delta(n(\delta + \xi)^{n-1}) \le \delta n(2N)^{n-1} < \epsilon. \end{aligned}$$

On the other hand, since $p \in A$ and $q \in \bar{A}$, we see $p^n \le x \le q^n$.

To sum up, we have showed that for each given $\epsilon > 0$, there are rational numbers p^n, q^n such that $p^n \le x, \xi^n \le q^n$ and $0 < q^n - p^n < \epsilon$. By the previous corollary, this shows $x = \xi^n$ for a positive real number ξ.

Finally, if $\eta > 0$ satisfying $\eta^n = x$ does not equal ξ, then either $\eta > \xi$ or $\eta < \xi$. Both are impossible because, in the first case, $\eta^n > \xi^n = x$ and in the second case, $\eta^n < x$. Thus, ξ is the unique positive real number satisfying $x = \xi^n$. □

In particular, $\sqrt{2}$ is not a rational number, but a real number. A real number that is not rational is called irrational.

Theorem 9 *If a and b are positive real numbers and n is a positive integer, then $(a \cdot b)^{1/n} = a^{1/n} \cdot b^{1/n}$. When there is no confusion, we omit the $\cdot$, and the above is written as $(ab)^{1/n} = a^{1/n}b^{1/n}$.*

Proof. By the previous theorem, we only need to check $\left(a^{\frac{1}{n}}b^{\frac{1}{n}}\right)^n = ab$. This is clearly true by the associative and commutative laws of multiplication. □

Example 12 *Let $b > 0$ be a real number, m, n be two positive integers. Show $(b^{\frac{1}{n}})^m = (b^m)^{\frac{1}{n}}$. We introduce notation*

$$b^{\frac{m}{n}} := \left(b^{\frac{1}{n}}\right)^m = (b^m)^{\frac{1}{n}}.$$

In an exercise, we show that if $m/n = p/q$ for positive integers m, n, p, q, then $b^{\frac{m}{n}} = b^{\frac{p}{q}}$. This makes the above notation made legitimately.

Proof. We only need to show $((b^{\frac{1}{n}})^m)^n = b^m$ as follows. By definition,

$$((b^{\frac{1}{n}})^m)^n = \underbrace{((b^{\frac{1}{n}})^m)((b^{\frac{1}{n}})^m)\dots((b^{\frac{1}{n}})^m)}_{n \text{ times}}$$

and

$$(b^{\frac{1}{n}})^m = \underbrace{(b^{\frac{1}{n}})(b^{\frac{1}{n}})\dots(b^{\frac{1}{n}})}_{m \text{ times}}.$$

Therefore,

$$\begin{aligned} ((b^{\frac{1}{n}})^m)^n &= \underbrace{\underbrace{(b^{\frac{1}{n}})(b^{\frac{1}{n}})\dots(b^{\frac{1}{n}})}_{m \text{ times}}\underbrace{(b^{\frac{1}{n}})(b^{\frac{1}{n}})\dots(b^{\frac{1}{n}})}_{m \text{ times}}\dots\underbrace{(b^{\frac{1}{n}})(b^{\frac{1}{n}})\dots(b^{\frac{1}{n}})}_{m \text{ times}}}_{n \text{ times}} \\ &= \underbrace{\underbrace{(b^{\frac{1}{n}})(b^{\frac{1}{n}})\dots(b^{\frac{1}{n}})}_{n \text{ times}}\underbrace{(b^{\frac{1}{n}})(b^{\frac{1}{n}})\dots(b^{\frac{1}{n}})}_{n \text{ times}}\dots\underbrace{(b^{\frac{1}{n}})(b^{\frac{1}{n}})\dots(b^{\frac{1}{n}})}_{n \text{ times}}}_{m \text{ times}} \\ &= b^m, \end{aligned}$$

where we used associative and commutative laws of multiplication. □

Example 13 *If $a, b > 0$ are positive real numbers, n is a positive integer, such that $a^n > b$, then $a > b^{\frac{1}{n}}$.*

Proof. It is impossible that $a = b^{\frac{1}{n}}$ as the latter term is the unique positive real solution to equation $x^n = b$. Assume for contradiction that $a < b^{\frac{1}{n}}$. Then

$$\begin{aligned} & a^n = (a)(a^{n-1}) < b^{\frac{1}{n}} a^{n-1} = b^{\frac{1}{n}}(a)(a^{n-2}) \\ < \; & b^{\frac{1}{n}} b^{\frac{1}{n}} a^{n-2} = b^{\frac{2}{n}} a^{n-2} = b^{\frac{2}{n}}(a)(a^{n-3}) \\ \vdots \; & \\ < \; & b^{\frac{n-1}{n}}(a) \\ < \; & b^{\frac{n-1}{n}} b^{\frac{1}{n}} = (b^{1/n})^{n-1}(b^{1/n}) = \left(b^{\frac{1}{n}}\right)^n = b, \end{aligned}$$

which is a contradiction because $a^n > b$. □

Note that the above proof also shows that

Lemma 2 *For positive real numbers b, c, and positive integer n, $c < b$ if and only if $c^{1/n} < b^{1/n}$.*

For the if part, this can be seen by putting $c^{1/n} = a$ in the blue part of the previous proof. The only if part follows from the previous example.

Example 14 *Let $A = \{x \in \mathbb{R} : \; x > 0, x^2 > 5\}$. Prove that A is non-empty, bounded below, and* $\inf A = \sqrt{5}$.

Proof. Clearly, $\sqrt{7} \in A$. Therefore, A is nonempty. All elements of A are positive, therefore A is bounded below. Consequently, the least upper bound property of $\mathbb{R}$ implies $\inf A \in \mathbb{R}$ exists.

For all $x \in A$, by the previous lemma, we have $x > \sqrt{5}$, therefore $\sqrt{5}$ is a lower bound of A.

It remains to show that the greatest lower bound of A is $\sqrt{5}$.

We argue by contradiction. Suppose $\sqrt{5} < \inf A$. Pick a real number a such that $\sqrt{5} < a < \inf A$ (which can be done by the denseness of $\mathbb{Q}$ in $\mathbb{R}$). Then a is an element of A (since a is a positive real number with $a^2 > 5$), which is smaller than the infimum of A, yielding a contradiction (because $\inf A$ is a lower bound). □

Intervals

There are certain special sets of real numbers that appear particularly often. These are called **intervals**, and take one of the following forms where a and b denote fixed real numbers:

$$(a, b) = \{x \in \mathbb{R} : a < x < b\}$$

$$[a, b) = \{x \in \mathbb{R} : a \leq x < b\}$$

$$(a, b] = \{x \in \mathbb{R} : a < x \leq b\}$$

$$[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$$

$$(a, \infty) = \{x \in \mathbb{R} : a < x\}$$

$$(-\infty, b) = \{x \in \mathbb{R} : x < b\}$$

$$[a, \infty) = \{x \in \mathbb{R} : a \leq x\}$$

$$(-\infty, b] = \{x \in \mathbb{R} : x \leq b\}$$

The sets with only round brackets are called **open intervals**, and the sets with only square brackets are called **closed intervals**.

Occasionally, the following notation is in use. We call $\mathbb{R} \cup \{\infty\} \cup \{-\infty\} =: [-\infty, \infty]$ the extended real line, and define for any $a, b \in \mathbb{R}$, $(a, \infty] := (a, \infty) \cup \{\infty\}$ and $[-\infty, b) := (-\infty, b) \cup \{-\infty\}$, and we understand $[a, \infty]$, $[-\infty, , b]$ likewise. We accept that $a < \infty$ and $b > -\infty$ for any real numbers a, b, and remark that ∞ and $-\infty$ are not real numbers.

Chapter 2

Sequences

2.1 Convergent sequences

A sequence say $\{a_n\}$ of real numbers is a function f, whose domain is $\mathbb{N}$ and whose codomain is $\mathbb{R}$ such that for each $n \in \mathbb{N}$, $f(n) = a_n$.

Other commonly used notation for a sequence $\{a_n\}$ is (a_n). We call a_n the nth term of the sequence. However, sometimes, when the context is clear, we may simply write a_n for the sequence $\{a_n\}$.

Definition 13 (Limit of a sequence) *A sequence $\{a_n\}$ converges to a number $l \in \mathbb{R}$ if for every $\epsilon > 0$, there exists a natural number N such that*

$$|a_n - l| < \epsilon$$

whenever $n \geq N$. In this case, we call the sequence $\{a_n\}$ convergent with the limit l, and write $a_n \to l$ as $n \to \infty$ or $\lim_{n\to\infty} a_n = l$.

- A sequence that fails to converge is said to diverge.

Example 15 *Assume that $a_n = C$ for all $n \in \mathbb{N}$, where C is a constant. Show that*

$$\lim_{n\to\infty} a_n = C.$$

Proof. Let $\epsilon > 0$ be fixed. We have

$$|a_n - C| = 0 < \epsilon, \quad \forall\ n \geq 1.$$

Therefore, for the given $\epsilon > 0$, we may take e.g., $N = 1$ in the definition of $\lim_{n\to\infty} a_n = C$. □

Example 16 *Prove that*

$$\lim_{n\to\infty} \frac{1}{n} = 0$$

from the definition.

Proof. Let $\epsilon > 0$ be fixed. Let $N = \left[\frac{1}{\epsilon}\right] + 1$. Here and below, for any real number x, $[x]$ denotes the largest integer that is smaller or equal to x.

Then for any $n \geq N$, we have

$$\left|\frac{1}{n} - 0\right| = \frac{1}{n} \leq \frac{1}{N} < \epsilon.$$

Thus we finish the proof by the definition. □

Example 17 *Show that*

$$\lim_{n\to\infty} \frac{1}{\ln n} = 0.$$

Proof. Let $\epsilon > 0$ be fixed. We take

$$N = \left[e^{\frac{1}{\epsilon}}\right] + 1.$$

When $n \geq N$, we have the chain of inequalities $n \geq N > e^{\frac{1}{\epsilon}}$, and thus

$$\frac{1}{\ln n} \leq \frac{1}{\ln N} < \frac{1}{\ln(e^{\frac{1}{\epsilon}})} = \epsilon.$$

The desired assertion follows from the definition. □

Example 18 *Show that the sequence* $\{(-1)^n\}$ *diverges.*

Proof. Suppose for contradiction that the sequence $\{(-1)^n\}$ converges to some real number A.

From the definition, there exists $N \in \mathbb{N}$, such that when $n > N$ we have

$$|(-1)^n - A| < 1.$$

By checking the above inequality against odd n and even n, one would obtain

$$|A + 1| < 1 \text{ and } |A - 1| < 1.$$

It then follows from the triangle inequality that

$$2 = |(A + 1) - (A - 1)| \leq |A + 1| + |A - 1| < 1 + 1 = 2,$$

which is a desired contradiction. □

2.2 Divergence to ∞ or $-\infty$

Definition 14 (Divergence to ∞) *Let* $\{a_n\}$ *be a sequence of real numbers. We say that* $\{a_n\}$ *diverges to* ∞ *and write* $\lim_{n\to\infty} a_n = \infty$ *or* $a_n \to \infty$ *as* $n \to \infty$*, if for every positive number* M *there is a natural number* N *such that* $x_n > M$ *for all* $n \geq N$*.*

Likewise, we say that $\{a_n\}$ diverges to $-\infty$ and write $\lim_{n\to\infty} a_n = -\infty$ or $a_n \to -\infty$ as $n \to \infty$, if for every negative number M there is a natural number N such that $x_n < M$ for all $n \geq N$.

Example 19 *Prove*

$$\frac{n^2 + 1}{n + 1} \to \infty \text{ as } n \to \infty$$

from the definition.

Proof. Let M be any positive number. Take the integer $N = [M] + 2$, then, for all $n \geq N$, we have

$$\frac{n^2 + 1}{n + 1} \geq \frac{n^2 - 1}{n + 1} = n - 1 \geq N - 1 = [M] + 1 > M.$$

Now the proof is completed. □

2.3 Properties of convergent sequences

2.3.1 Uniqueness

Theorem 10 (Uniqueness) *If*

$$\lim_{n\to\infty} x_n = A \text{ and } \lim_{n\to\infty} x_n = B$$

for some real numbers A, B*, then* $A = B$*.*

Proof. Let $\epsilon > 0$ be fixed. Then, by definition, we can find a natural number N_1 so that

$$|x_n - A| < \epsilon/2$$

for $n \geq N_1$. We can also find a natural number N_2 so that

$$|x_n - B| < \epsilon/2$$

for $n \geq N_2$. Let $N_3 = \max\{N_1, N_2\}$. Then both inequalities

$$|x_{N_3} - A| < \epsilon/2 \text{ and } |x_{N_3} - B| < \epsilon/2$$

hold.

It follows that

$$|A - B| \leq |x_{N_3} - A| + |x_{N_3} - B| < \epsilon.$$

Since $\epsilon > 0$ was arbitrarily fixed, it follows that $A = B$. □

2.3.2 Boundedness

Definition 15 (Bounded sequence) *A sequence $\{a_n\}$ is bounded if there is a real number $M > 0$ such that*

$$|a_n| \leq M \ \forall \ n \in \mathbb{N}.$$

Theorem 11 (Boundedness property) *Every convergent sequence is bounded.*

Proof. Suppose that $a_n \to A$ as $n \to \infty$. Then there exists a natural number N so that

$$|a_n - A| \leq 1$$

whenever $n \geq N$. By triangle inequality we have

$$|a_n| = |a_n - A + A| \leq |a_n - A| + |A| \leq |A| + 1$$

for all $n \geq N$.

Now we define

$$M := \max\{|a_1|, |a_2|, \cdots, |a_{N-1}|, |A| + 1\}.$$

Then $|a_n| \leq M$ for all $n \in \mathbb{N}$. □

An immediate consequence of the previous statement is the following one.

Corollary 2 *An unbounded sequence must diverge.*

Example 20 *Show the sequence $\{s_n\}$ defined by*

$$s_n = \sum_{k=1}^{n} \frac{1}{k} \ \ \forall \ n \in \mathbb{N}$$

diverges.

Proof. Examine the terms of the sequence with index $1 = 2^0, 2 = 2^1, 4 = 2^2, 8 = 2^3, \cdots$:

$$s_1 = 1$$

$$s_2 = 1 + \frac{1}{2}$$

$$s_4 = 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) \geq 1 + \frac{1}{2} + 2\left(\frac{1}{4}\right)$$

$$s_8 = 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) \geq 1 + \frac{1}{2} + 2\left(\frac{1}{4}\right) + 4\left(\frac{1}{8}\right)$$

and in general we have that

$$s_{2^n} = 1 + \sum_{j=1}^{n} \sum_{k=2^{j-1}+1}^{2^j} \frac{1}{k} \geq 1 + \frac{n}{2}$$

for all $n \in \mathbb{N}$, where the last inequality holds because

$$\sum_{k=2^{n-1}+1}^{2^n} \frac{1}{k} \geq \sum_{k=2^{n-1}+1}^{2^n} \frac{1}{2^n} = \frac{2^n - 2^{n-1}}{2^n} = \frac{1}{2}.$$

Thus the sequence is unbounded and thus divergent. □

2.3.3 Arithmetic operations

Theorem 12 *Suppose that $\{x_n\}$ and $\{y_n\}$ are convergent sequences and C is a real number.*

(a) $\lim_{n\to\infty}(x_n + y_n) = \lim_{n\to\infty} x_n + \lim_{n\to\infty} y_n$.

(b) $\lim_{n\to\infty}(x_n y_n) = (\lim_{n\to\infty} x_n) \times (\lim_{n\to\infty} y_n)$. *In particularly, we have, for any constant C,*

$$\lim_{n\to\infty}(Cx_n) = C\left(\lim_{n\to\infty} x_n\right)$$

and

$$\lim_{n\to\infty}(x_n - y_n) = \lim_{n\to\infty} x_n - \lim_{n\to\infty} y_n.$$

(c) If $y_n \neq 0$ for all $n \in \mathbb{N}$ and $\lim_{n\to\infty} y_n \neq 0$, then

$$\lim_{n\to\infty}\left(\frac{x_n}{y_n}\right) = \frac{\lim_{n\to\infty} x_n}{\lim_{n\to\infty} y_n}.$$

Proof. We only prove (b), and leave the proofs of the other assertions as an exercise. Suppose $x_n \to A$, $y_n \to B$ as $n \to \infty$, and $M > 0$ is an upper bound for $\{|y_n|\}$. Let $\epsilon > 0$ be fixed. By definition, there exists $N \in \mathbb{N}$ such that both

$$|x_n - A| < \frac{\epsilon}{2M}$$

and

$$|y_n - B| < \frac{\epsilon}{2|A| + 1}$$

hold for all $n \geq N$. Then we have

$$\begin{aligned} |x_n y_n - AB| &= |x_n y_n - A y_n + A y_n - AB| \\ &\leq |x_n - A||y_n| + |A||y_n - B| \\ &\leq |x_n - A|M + |A||y_n - B| \\ &\leq \frac{\epsilon}{2M} \times M + \frac{\epsilon}{2|A|+1} \times |A| \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon. \end{aligned}$$

Now (b) follows. □

Example 21 *Find the limit of the sequence* $\left\{\frac{3n^2+2n+9}{6n^2+4n+6}\right\}$.

Solution. We have

$$\frac{3n^2 + 2n + 9}{6n^2 + 4n + 6} = \frac{3 + \frac{2}{n} + \frac{9}{n^2}}{6 + \frac{4}{n} + \frac{6}{n^2}} \to \frac{3 + 2 \cdot 0 + 9 \cdot 0}{6 + 4 \cdot 0 + 6 \cdot 0} = \frac{1}{2}$$

as $n \to \infty$.

2.4 Sandwich theorem

Lemma 3 *Suppose that $\{a_n\}$ is convergent sequence and $a_n \geq 0$ for all n. Then*

$$\lim_{n\to\infty} a_n \geq 0.$$

Proof. Set $A = \lim_{n\to\infty} a_n$. Let $\epsilon > 0$ be fixed. There exists $N \in \mathbb{N}$ such that for all $n \geq N$ we have

$$|a_n - A| < \epsilon.$$

From the above we obtain that for all $n \geq N$,

$$A > a_n - \epsilon \geq -\epsilon.$$

Since $\epsilon > 0$ was arbitrarily fixed, we see that $A \geq 0$. □

Corollary 3 *Suppose that $\{x_n\}$ and $\{y_n\}$ are convergent sequences and $x_n \leq y_n$ for all n. Then*

$$\lim_{n\to\infty} x_n \leq \lim_{n\to\infty} y_n.$$

Proof. Apply the previous lemma to $a_n := y_n - x_n$. □

The next result is referred to as the sandwich theorem or squeeze theorem.

Theorem 13 (Sandwich/Squeeze) *Suppose that $\{x_n\}$ and $\{y_n\}$ are convergent sequences such that $\lim_{n\to\infty} x_n = \lim_{n\to\infty} y_n$ and $x_n \leq z_n \leq y_n$ for all n. Then $\{z_n\}$ converges, and*

$$\lim_{n\to\infty} x_n = \lim_{n\to\infty} z_n = \lim_{n\to\infty} y_n.$$

Proof. Let $A := \lim_{n\to\infty} x_n = \lim_{n\to\infty} y_n$. It follows that for any $\epsilon > 0$, there exist some $N_1, N_2 \in \mathbb{N}$ such that

$$-\epsilon < x_n - A < \epsilon, \forall\ n \geq N_1;$$

and

$$-\epsilon < y_n - A < \epsilon, \forall\ n \geq N_2.$$

Set $N = \max\{N_1, N_2\}$, then both of the above inequalities hold for all $n \geq N$. Since $x_n \leq z_n \leq y_n$ we conclude that

$$-\epsilon < x_n - A \leq z_n - A \leq y_n - A < \epsilon, \forall\ n \geq N.$$

It follows that $-\epsilon < z_n - A < \epsilon, \forall\ n \geq N$. □

Example 22 *Let θ be some real number. Find the limit of the sequence $\{x_n\}$, where*

$$x_n = \frac{\sin n\theta}{n}.$$

Solution. It holds that

$$-\frac{1}{n} \leq \frac{\sin n\theta}{n} \leq \frac{1}{n}$$

for all $n \in \mathbb{N}$. Both the outer sequences converge to 0 and hence the inner sequence must converge to 0 by the sandwich theorem.

Example 23 *Find the limit* $\lim_{n\to\infty} \frac{2^n}{n!}$.

Solution. Let $x_n = \frac{2^n}{n!}$. We have

$$0 \leq x_n \leq \frac{4}{n}.$$

Since $\lim_{n\to\infty} \frac{4}{n} = 0$ we have $\lim_{n\to\infty} \frac{2^n}{n!} = 0$ by the squeeze theorem.

Example 24 *Let*

$$x_n = \frac{n!}{n^n}.$$

Find $\lim_{n\to\infty} x_n$.

Solution. It is easy to see that

$$0 < x_n \le \frac{1}{n}.$$

By the squeeze theorem, $\lim_{n\to\infty} x_n = 0$.

Example 25 *Let*

$$x_n = \sqrt[n]{n}.$$

Find the limit $\lim\limits_{n\to\infty} x_n$.

Solution. Let $x_n = 1 + h_n$, where $h_n > 0$ for $n > 1$. We have, for $n > 1$

$$n = (1 + h_n)^n \ge \frac{n(n-1)}{2} h_n^2.$$

It follows that

$$0 < h_n < \sqrt{\frac{2}{n-1}}, \quad \text{for } n > 1.$$

Thus we have

$$1 \le x_n = 1 + h_n \le 1 + \sqrt{\frac{2}{n-1}}.$$

Since $\lim\limits_{n\to\infty} \left(1 + \sqrt{\frac{2}{n-1}}\right) = 1$ we have $\lim\limits_{n\to\infty} x_n = 1$ by the squeeze theorem.

Using the squeeze theorem and the above example, one can show that $\lim_{n\to\infty} a^{1/n} = 1$ for all $a > 0$.

2.5 Some frequently encountered null sequences

A sequence $\{a_n\}$ of real numbers is called a null sequence if $a_n \to 0$ as $n \to \infty$.

Example 26 (Common null sequences) *Prove the following assertions.*

(i) If $s > 0$ *then* $\frac{1}{n^s} \to 0$ *as* $n \to \infty$.

(ii) If $|\lambda| < 1$ *then* $\lambda^n \to 0$ *as* $n \to \infty$.

(iii) If $s \in \mathbb{R}$ *and* $|\lambda| > 1$ *then* $\frac{n^s}{\lambda^n} \to 0$ *as* $n \to \infty$.

(iv) If $s \in \mathbb{R}$ *then* $\frac{n^s}{n!} \to 0$ *as* $n \to \infty$.

(v) If $\lambda \in \mathbb{R}$ *then* $\frac{\lambda^n}{n!} \to 0$ *as* $n \to \infty$.

(vi) We have $\frac{n!}{n^n} \to 0$ *as* $n \to \infty$.

Proof. Only (iii) and (iv) need proofs, (i) and (ii) are simple, and we skip them. (vi) has been proved in a previous example. If (iii) and (v) are proved, then they imply (iv) by the algebra of limits. It is better to prove (iii) after we covered the monotone convergence theorem. Let us prove (v) here.

We need to show $|\frac{\lambda^n}{n!}| = \frac{|\lambda|^n}{n!} \to 0$ as $n \to \infty$. Let $\mu = |\lambda|$ and $a_n = \frac{\mu^n}{n!}$. Then

$$a_{n+1} = a_n \frac{\mu}{n+1}.$$

Let $N > 0$ be a natural number such that $0 \le \frac{\mu}{N+1} < 1$. Then

$$0 \le a_{N+k} \le a_N \left(\frac{\mu}{N+1}\right)^k$$

for all natural numbers $k \ge 1$. Since $(\frac{\mu}{N+1})^k \to 0$ as $k \to \infty$ (by (ii)), we see that $a_{N+k} \to 0$ as $k \to \infty$. But this also implies $\{a_n\}$ converges to 0 as $n \to \infty$. □

The above example asserts that the factorial growth ($n!$) beats the exponential growth (λ^n for some $\lambda > 0$), which beats the polynomial growth (n^s for some $s > 0$).

2.6 Monotone sequences

Definition 16 *A sequence $\{x_n\}$ is (strictly) increasing if $x_n < x_{n+1}$ holds for all $n \in \mathbb{N}$. A sequence $\{y_n\}$ is (strictly) decreasing if $y_n > y_{n+1}$ holds for all $n \in \mathbb{N}$.*

Definition 17 *A sequence $\{x_n\}$ is non-decreasing if $x_n \leq x_{n+1}$ holds for all $n \in \mathbb{N}$. A sequence $\{y_n\}$ is non-increasing if $y_n \geq y_{n+1}$ holds for all $n \in \mathbb{N}$.*

Non-increasing sequences and non-decreasing sequences are both referred to as monotone sequences.

Theorem 14 (Monotone convergence theorem) *Suppose that $\{x_n\}$ is a monotone sequence. Then, $\{x_n\}$ is convergent if and only if $\{x_n\}$ is bounded.*

Proof. Assume that $\{x_n\}$ is non-decreasing. (If it is non-increasing, set $y_n = -x_n$ and work with y_n instead.) The 'only if' part follows form Theorem 11. Now we turn to the 'if' part. Since $\{x_n\}$ is bounded, thus its supremum exists. Set

$$L := \sup\{x_n : n \in \mathbb{N}\}.$$

For any $\epsilon > 0$, since L is the least upper bound, $L - \epsilon$ is not an upper bound for the set $\{x_n : n \in \mathbb{N}\}$. It follows that there exists an $N \in \mathbb{N}$, such that $x_N > L - \epsilon$. Due to the fact that the sequence $\{x_n\}$ is non-decreasing, we have

$$L - \epsilon < x_N \leq x_n \leq L < L + \epsilon$$

for all $n \geq N$. Put it differently,

$$L - \epsilon < x_n < L + \epsilon$$

for all $n \geq N$. Thus the theorem is proved. □

Example 27 *Show that if $s \in \mathbb{R}$ and $|\lambda| > 1$ then $\frac{n^s}{\lambda^n} \to 0$ as $n \to \infty$.*

Proof. We need prove that $|\frac{n^s}{\lambda^n} - 0| = \frac{n^s}{|\lambda|^n} \to 0$ as $n \to \infty$. It is sufficient to prove this for M with M being a positive integer, bigger than s.

Let $\mu = |\lambda| > 1$ and $a_n := \frac{n^M}{\mu^n}$.

Firstly show that for some N, $\{a_n\}$ is decreasing in $n > N$. For this purpose, check

$$\frac{a_{n+1}}{a_n} = \frac{1}{\mu}\left(\frac{n+1}{n}\right)^M = \frac{1}{\mu}(1 + \frac{1}{n})^M.$$

Since $1 + \frac{1}{n} \to 1$, so does $(1 + \frac{1}{n})^M$, as $n \to \infty$. Consequently, there is some natural number N such that $(1 + \frac{1}{n})^M < \mu$ for all $n > N$, because $\mu > 1$. Now for all $n > N$,

$$0 < \frac{a_{n+1}}{a_n} < \frac{1}{\mu}\mu = 1.$$

Therefore, $\{a_n\}_{n>N}$ is a decreasing sequence. Clearly, $\{a_n\}_{n>N}$ is bounded below by 0. It is also bounded from above by e.g., a_{N+1}. So we may apply the monotone convergence theorem for that $a_n \to a$ for some $a \in \mathbb{R}$ as $n \to \infty$.

Next, let us compute this $a \in \mathbb{R}$.

Note $a_{n+1} = \frac{(n+1)^M}{\mu^{n+1}} = \frac{1}{\mu}\frac{(n+1)^M}{n^M}\frac{n^M}{\mu^n} = \frac{1}{\mu}\frac{(n+1)^M}{n^M}a_n$.

By the algebra of limits, we see from the above equalities that $a = \frac{1}{\mu}a$. This implies $a = 0$. □

2.7 Natural exponent e

Theorem 15 *The sequence $\{x_n\}$ defined by*

$$x_n = \left(1 + \frac{1}{n}\right)^n \quad \forall\ n \in \mathbb{N}$$

is convergent.

Proof. In this proof, we shall apply the following Bernoulli inequality:

$$(1+x)^n \geq 1+nx, \quad \forall n \in \mathbb{N} \text{ and } x \geq -1.$$

This inequality can be proved by induction, and we leave it as an exercise.

For any $n \in \mathbb{N}$, by Bernoulli inequality,

$$\begin{aligned}\frac{x_{n+1}}{x_n} &= \frac{n^n(n+2)^{n+1}}{(n+1)^{2n+1}} = \left(\frac{n^2+2n}{(n+1)^2}\right)^n \frac{n+2}{n+1} \\ &= \left(1-\frac{1}{n^2+2n+1}\right)^n \frac{n+2}{n+1} \\ &\geq \left(1-\frac{n}{n^2+2n+1}\right)\frac{n+2}{n+1} \\ &= \frac{n^3+3n^2+3n+2}{n^3+3n^2+3n+1} > 1.\end{aligned}$$

Set

$$y_n = \left(1+\frac{1}{n}\right)^{n+1}$$

for all $n \in \mathbb{N}$. Similarly to the above, for any positive integer $n \geq 2$, by Bernoulli inequality again,

$$\begin{aligned}\frac{y_{n-1}}{y_n} &= \frac{n^{2n+1}}{(n-1)^n(n+1)^{n+1}} = \left(\frac{n^2}{n^2-1}\right)^n \frac{n}{n+1} \\ &= \left(1+\frac{1}{n^2-1}\right)^n \frac{n}{n+1} \\ &\geq \left(1+\frac{n}{n^2-1}\right)\frac{n}{n+1} \\ &= \frac{n^3+n^2-n}{n^3+n^2-n-1} > 1.\end{aligned}$$

It follows that $\{x_n\}$ and $\{y_n\}$ are increasing and decreasing respectively. Furthermore, for any $n \in \mathbb{N}$, we have

$$2 = x_1 \leq x_n \leq y_n \leq y_1 = 4.$$

Thus we conclude that both $\{x_n\}$ and $\{y_n\}$ are bounded. By the monotone convergence theorem, we see that both $\{x_n\}$ and $\{y_n\}$ are convergent. □

Note that in the above proof, since $y_n = x_n(1+\frac{1}{n})$, by taking $n \to \infty$ on both sides, it follows that both sequences $\{x_n\}$ and $\{y_n\}$ converge to the same limit.

Definition 18 (Natural exponent e) *The limit $e := \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n$ is called the natural exponent (or exponential constant).*

One can give the following interpretation of the constant e.

An account starts with £1.00 and pays 100% interest per year. If the interest is credited once, at the end of the year, the value of the account at year-end will be £2.00. What happens if the interest is computed and credited more frequently during the year?

If the interest is credited twice in the year, the interest rate for each 6 months will be 50%, so the initial 1 is multiplied by 1.5 twice, yielding $£1.00 \times 1.5^2 = 2.25$ at the end of the year. Compounding quarterly yields $£1.00 \times 1.25^4 = 2.4414\ldots$, and compounding monthly yields $£1.00 \times (1+1/12)^{12} = 2.613035\ldots$ If there are n compounding intervals, the interest for each interval will be $100\%/n$ and the value at the end of the year will be $£1.00 \times (1+1/n)^n$. Thus, $£e$ is the amount you will receive in the case of the so called continuous compounding.

From the proof of Theorem 15, we deduce the following inequality.

Lemma 4 *For $n \in \mathbb{N}$, $\frac{1}{n+1} < \ln(1+\frac{1}{n}) < \frac{1}{n}$.*

Proof. Inspecting the proof of Theorem 15 reveals that $(1+\frac{1}{n})^n < e < (1+\frac{1}{n})^{n+1}$. Hence, $n\ln(1+\frac{1}{n}) < 1 < (n+1)\ln(1+\frac{1}{n})$. Dividing the first inequality by n and the second inequality by $n+1$, we see $\frac{1}{n+1} < \ln(1+\frac{1}{n}) < \frac{1}{n}$. □

2.8 Stolz theorem

We introduce a theorem of the Austrian mathematician Stolz, which is useful for computing the limit of certain convergent sequences.

Theorem 16 (Stolz theorem) *Suppose (y_n) is strictly increasing and diverges to ∞. Let (x_n) be another sequence in $\mathbb{R}$. Suppose $\lim_{n\to\infty} \frac{x_{n+1}-x_n}{y_{n+1}-y_n} = l \in \mathbb{R}$, then $\frac{x_n}{y_n} \to l$ as $n \to \infty$.*

Proof. Let $\epsilon > 0$ be fixed. From the assumption that $\lim_{n\to\infty} \frac{x_{n+1}-x_n}{y_{n+1}-y_n} = l$ and $y_n \to \infty$ as $n \to \infty$, we see that there is some natural number $N \geq 1$ satisfying

$$l - \epsilon/3 < \frac{x_{n+1} - x_n}{y_{n+1} - y_n} < l + \epsilon/3$$

and $y_n > 0$ for all $n \geq N$.

From $(x_{n+1} - x_n) < (y_{n+1} - y_n)(l + \epsilon/3)$ for all $n \geq N$, we see $x_{n+1} - x_N < (y_{n+1} - y_N)(l + \epsilon/3)$ for all $n \geq N$. Similarly, $x_{n+1} - x_N > (y_{n+1} - y_N)(l - \epsilon/3)$ for all $n \geq N$.

Now,

$$x_{n+1} - x_N < (l + \epsilon/3)(y_{n+1} - y_N)$$

implies $x_{n+1} < x_N + (l + \epsilon/3)y_{n+1} - (l + \epsilon/3)y_N$ and so $\frac{x_{n+1}}{y_{n+1}} < \frac{x_N}{y_{n+1}} + (l + \epsilon/3) - (l + \epsilon/3)\frac{y_N}{y_{n+1}}$ and similarly, $\frac{x_{n+1}}{y_{n+1}} > \frac{x_N}{y_{n+1}} + (l - \epsilon/3) - (l - \epsilon/3)\frac{y_N}{y_{n+1}}$, for all $n \geq N$.

Since $y_n \to \infty$ as $n \to \infty$, there is some natural number $N_1 \geq N$ such that $\frac{x_{n+1}}{y_{n+1}} < \frac{x_N}{y_{n+1}} + (l + \epsilon/3) - (l + \epsilon/3)\frac{y_N}{y_{n+1}} < l + \epsilon$ for all $n \geq N_1$.

Similarly, there is some natural number $N_2 \geq N$ such that $\frac{x_{n+1}}{y_{n+1}} > \frac{x_N}{y_{n+1}} + (l - \epsilon/3) - (l - \epsilon/3)\frac{y_N}{y_{n+1}} > l - \epsilon$ for all $n \geq N_2$.

That means, for all $n \geq \max\{N_1, N_2\}$, $|\frac{x_{n+1}}{y_{n+1}} - l| < \epsilon$. This completes the proof. □

The special case of the Stolz theorem when $y_n = n$ was obtained by Cauchy.

With the Stolz theorem, it is easy to see that if $a_n \to a \in \mathbb{R}$, then $\frac{\sum_{k=1}^n a_k}{n} \to a$ as $n \to \infty$, too. This fact is ascribed to Cauchy.

Another useful example is as follows.

Example 28 *For each $\alpha > 0$, show that $\lim_{n\to\infty} \frac{(\ln n)^\alpha}{n} = 0$.*

Proof. It is sufficient to show the statement for $\alpha = k$ for some $k \in \mathbb{N}$ in view of the sandwich theorem. In view of the Stolz theorem, we consider

$$\begin{aligned}
0 &\leq (\ln(n+1))^k - \ln(n)^k = (\ln(n+1) - \ln n)\left\{(\ln(n+1))^{k-1} + (\ln(n+1))^{k-2}\ln n + \cdots + (\ln n)^{k-1}\right\} \\
&= \ln(1 + \frac{1}{n})\left\{(\ln(n+1))^{k-1} + (\ln(n+1))^{k-2}\ln n + \cdots + (\ln n)^{k-1}\right\} \\
&\leq \frac{1}{n}k(\ln(n+1))^{k-1} \\
&= \frac{1}{n}k(\ln(n+1) - \ln n + \ln n)^{k-1} \\
&\leq \frac{1}{n}k(\frac{1}{n} + \ln n)^{k-1} \\
&= k\sum_{i=0}^{k-1}\binom{k-1}{i}\frac{1}{n^i}\frac{(\ln n)^{k-1-i}}{n},
\end{aligned}$$

where Lemma 4 asserting $\ln(1 + \frac{1}{n}) < \frac{1}{n}$ was in use.

When $k = 1$, the above inequalities how that $(\ln(n+1))^k - \ln(n)^k \to 0$ as $n \to \infty$, and by the Stolz theorem, we see that $\frac{(\ln n)^k}{n} \to 0$ as $n \to \infty$, when $k = 1$.

Assume that $\lim_{n\to\infty} \frac{(\ln n)^k}{n} = 0$ for some $k \in \mathbb{N}$. Then, by the Stolz theorem, the previous inequalities (with k being replaced by $k + 1$) imply that $\lim_{n\to\infty} \frac{(\ln n)^{k+1}}{n} = 0$, too. The statement follows now, by induction. □

2.9 Subsequences

Definition 19 (Subsequence) *Let $\{x_n\}$ be a sequence. Then a subsequence of this sequence is a sequence $\{x_{n_i}\}_{i=1}^{\infty}$ with $\{n_i\}$ being a strictly increasing sequence of positive integers.*

It is useful to note the following: if $\{x_{n_k}\}$ is a subsequence of $\{x_n\}$, then $n_k \geq k$ for each $k \in \mathbb{N}$.

Example 29 *Consider the sequence $\{x_n\} = \{(-1)^n\}$. We can take $n_i = 2i$ for all $i \in \mathbb{N}$, and form the corresponding subsequence $\{x_{n_i}\}$ given by $x_{n_i} = 1$ for all $i \in \mathbb{N}$.*

Example 30 *The unbounded sequence $\{x_n\}$ with*

$$x_n = \begin{cases} 0, & \text{for odd } n; \\ n, & \text{for even } n \end{cases}$$

has a subsequence $\{x_{2k-1}\}_{k=1}^{\infty}$ with $n_k = 2k-1$ and $x_{n_k} = 0$ for all $k \in \mathbb{N}$.

Theorem 17 *If $\{x_n\}$ is a sequence of real numbers converging to $l \in \mathbb{R}$, then every subsequence of $\{x_n\}$ converges to l.*

Proof. Let $\epsilon > 0$ be fixed. Since $\{x_n\}$ converges to l, there exists $N \in \mathbb{N}$ such that

$$|x_n - l| < \epsilon \quad \text{for all } n \geq N.$$

Let $\{x_{n_k}\}$ be a subsequence of $\{x_n\}$. Hence, if $k \geq N$ then $n_k \geq N$. With this observation we obtain

$$|x_{n_k} - l| < \epsilon \quad \text{for all } k \geq N.$$

Thus, we finished the proof. □

A similar argument shows that if $\{x_n\}$ diverges to ∞ or $-\infty$, then every subsequence of $\{x_n\}$ diverges to ∞ or $-\infty$, respectively. For example, if $\{x_n\}$ diverges to ∞, and $\{x_{n_k}\}$ is a subsequence of $\{x_n\}$, then for each $M > 0$, there is some $N \in \mathbb{N}$ such that $n \geq N$ implies $x_n > M$. But then for each $k \geq N$, it holds that $n_k \geq k \geq N$ and thus $x_{n_k} > M$.

In view of the previous theorem, we can conclude that a sequence $\{x_n\}$ does not converge, once we find two subsequences of $\{x_n\}$, which converge to distinct limits.

Example 31 *Prove that the sequence $\{a_n\}$ given by $a_n = (-1)^n$ does not converge, by using the above observation.*

Proof. Consider the subsequences $\{a_{2n}\}$ and $\{a_{2n-1}\}$. Since $a_{2n} = 1$ for all $n \in \mathbb{N}$, we have that $a_{2n} \to 1$ as $n \to \infty$. Since $a_{2n-1} = -1$ for all $n \in \mathbb{N}$, we have that $a_{2n} \to -1$ as $n \to \infty$. Since $1 \neq -1$, we conclude that $\{a_n\}$ does not converge. □

2.10 Bolzano-Weierstrass theorem

Given a sequence of real numbers, can we always select a subsequence that is monotone? The answer to this question is affirmative.

Theorem 18 *Every sequence of real numbers has a monotone subsequence.*

Proof. Consider the set

$$S := \{n \in \mathbb{N} : a_m \leq a_n \quad \text{for all } m > n\}.$$

Since we do not know whether it is a finite set or an infinite set, let us consider these two possibilities in turn.

First assume that S is $\emptyset$ or finite. Let n_1 be any natural number that is larger than all of the elements of S (since S is finite). Then we have $n_1 \notin S$. Hence, by the definition of S, there exists $n_2 > n_1$ such that $a_{n_2} > a_{n_1}$. Similarly, since $n_2 \notin S$ there exists $n_3 > n_2$ such that $a_{n_3} > a_{n_2}$. Continuing in this way we obtain an increasing subsequence $\{a_{n_k}\}$, as desired.

Now we turn to the case of infinite S. We may write $S := \{n_k : k \in \mathbb{N}\}$ where $\{n_k\}$ is a strictly increasing sequence of natural numbers. We claim that the sequence $\{a_{n_k}\}$ is non-increasing. To prove this, let $k \in \mathbb{N}$ be arbitrarily fixed, and observe that $a_m \leq a_{n_k}$ for all $m > n_k$. Since $n_{k+1} > n_k$, in particular we have that $a_{n_{k+1}} \leq a_{n_k}$. Thus $\{a_{n_k}\}$ is non-increasing. □

The next result is after the Czech mathematician Bolzano and the German mathematician Weierstrass.

Theorem 19 (Bolzano-Weierstrass theorem) *Every bounded sequence has a convergent subsequence.*

Proof. In the previous theorem, we know that any sequence has a monotone subsequence. Since by assumption the given sequence is bounded, the corresponding monotone subsequence is also bounded, which thus converges by the monotone convergence theorem. □

2.11 Bolzano-Cauchy convergence criterion

Theorem 20 (Bolzano-Cauchy) *A sequence $\{x_n\}$ of real numbers is convergent if and only if for each $\epsilon > 0$, there exists an natural number N such that*

$$|x_n - x_m| < \epsilon$$

for all $n \geq N$ and $m \geq N$.

A sequence $\{x_n\}$ satisfying the condition in Theorem 20 is called fundamental or Cauchy.

This theorem is ascribed to the Czech mathematician Bolzano and the French mathematician Cauchy. Often enough, it was simply referred to as the Cauchy criterion.

The main advantage of the Bolzano-Cauchy criterion is that one can apply it to show a sequence converges without knowing the value of the limit in advance.

The proof of this theorem appeals to the Bolzano-Weierstrass theorem.

Proof of Theorem 20. First, let us prove that, if (x_n) converges, then it is Cauchy, as follows.

Let $\epsilon > 0$ be fixed. $x_n \to A \in \mathbb{R}$ implies, for some natural number $N \geq 1$, $|x_n - A| < \frac{\epsilon}{2}$ for all $n \geq N$.

Now for each $n, m \geq N$,

$$|x_n - x_m| = |x_n - A + A - x_m| \leq |x_n - A| + |x_m - A| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

This shows that (x_n) is Cauchy.

Second, let us prove that, if (x_n) is Cauchy, then it is convergent, as follows.

Note, if (x_n) is Cauchy, then it is bounded. Indeed, for a given $\epsilon > 0$, (x_n) being Cauchy implies that for some natural number $N \geq 1$, $|x_n - x_m| < \epsilon$ for all $m, n \geq N$. In particular, $|x_n - x_N| < \epsilon$ for all $n \geq N$.

Now for $n \geq N$, $x_N - \epsilon < x_n < x_N + \epsilon$. As a result, for all $n \geq 1$,

$$\min\{x_N - \epsilon, x_1, \ldots, x_{N-1}\} \leq x_n \leq \max\{x_N + \epsilon, x_1, \ldots, x_{N-1}\}.$$

This shows that (x_n) is a bounded sequence.

Next, the boundedness of (x_n) implies that it has a convergent subsequence (x_{n_k}) by the Bolzano-Weierstrass theorem. Let us say $x_{n_k} \to A$ as $k \to \infty$. Now we show that $x_n \to A$, too.

Let some $\epsilon > 0$ be fixed. Then (x_n) being Cauchy implies that for some natural number $N \geq 1$, $|x_n - x_m| < \frac{\epsilon}{2}$ for all $m, n \geq N$.

(x_{n_k}) converging to A implies that for some natural number $n_K \geq 1$, $|x_{n_k} - A| < \frac{\epsilon}{2}$ for all $n_k \geq n_K$ (i.e., for all $k \geq K$).

Now take some $n_k \geq \max\{n_K, N\}$. It holds that

$$|x_n - A| = |x_n - x_{n_k} + x_{n_k} - A| \leq |x_n - x_{n_k}| + |x_{n_k} - A| < |x_n - x_{n_k}| + \frac{\epsilon}{2}.$$

Finally, for all $n \geq N$, $|x_n - A| < |x_n - x_{n_k}| + \frac{\epsilon}{2} < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$, as required. □

2.12 Upper limit and lower limit

Consider a sequence $\{a_n\}$ of real numbers.

We say $\lim_{n\to\infty} a_n$ exists (or is defined) if either $\{a_n\}$ converges, or it diverges to ∞ or $-\infty$. In the latter cases, we write $\lim_{n\to\infty} a_n = \infty$ or $-\infty$, respectively. So now, the limit of a sequence is allowed to be $\infty, -\infty$ or a real number.

Then consider all the subsequences of $\{a_n\}$, whose limit exists. The maximal one of all such limits is called the upper limit of $\{a_n\}$, and is denoted by

$$\limsup_{n\to\infty} a_n;$$

the minimal one of all such limits is called the lower limit of $\{a_n\}$, and is denoted by

$$\liminf_{n\to\infty} a_n.$$

The above definition needs some clarifications, because first, one should accept that $-\infty < x < \infty$ for all real numbers x, and second, one must be assured that the maximal one and the minimal one exist in the above definitions. If they do, then we say the upper limit and the lower limit exist. This is indeed confirmed in the following result.

Theorem 21 *For any sequence $\{a_n\}$ of real numbers, $\limsup_{n\to\infty} a_n$ and $\liminf_{n\to\infty} a_n$ exist.*

Proof. We only show that $\limsup_{n\to\infty} a_n$ exists. The case of $\liminf_{n\to\infty} a_n$ can be proved in a similar manner.

Step 1. If $\{a_n\}$ is unbounded from above, then there is some $n_1 \geq 1$ such that $a_{n_1} > 1$. Since $\{a_n\}_{n>n_1}$ is still unbounded from above, 2 is not its upper bound, and one can find some $n_2 > n_1$ such that $a_{n_2} > 2$. An inductive argument shows that there is a strictly increasing sequence $\{n_k\}_{k\geq 1}$ such that $a_{n_k} > k$ for all $k \geq 1$. Then $\lim_{k\to\infty} a_{n_k} = \infty$. It follows that $\limsup_{n\to\infty} a_n$ exists and equals ∞.

Step 2. Assume now that $\{a_n\}$ is bounded from above. Consider $M_n = \sup_{k\geq n}\{a_k\}$. Since $\{a_n\}$ is bounded above, $M_n < \infty$ exists by the least upper bound property of the set of real numbers.

Evidently, $\{M_n\}_{n\geq 1}$ is a nonincreasing sequence. If it is bounded from below, then it converges to some real number say M. If it is unbounded from below, then $\{M_n\}$ diverges to $M = -\infty$. The reason is as follows. Let $C < 0$ be fixed. It cannot be a lower bound for $\{M_n\}$. Therefore, there is some natural number N such that $M_N < C$. Since $\{M_n\}$ is nonincreasing, for all $n \geq N$, $M_n < C$.

In case $M = -\infty$, then for any $-\epsilon < 0$, there is some $N \geq 1$ such that $M_n < -\epsilon$ for all $n \geq N$, so that $a_n < -\epsilon$ for all $n \geq N$. This implies $\lim_{n\to\infty} a_n = -\infty$, and consequently, every subsequence of $\{a_n\}$ diverges to $-\infty$, as was observed earlier. Thus, $\limsup_{n\to\infty} a_n$ exists and equals $-\infty$.

It remains to consider the case of M being a real number. We first show that M is the limit of some subsequence of $\{a_n\}$. The construction is based on the next two observations.

Observation 1: For any $\epsilon > 0$, there is some natural number N such that $M_n < M + \epsilon$ for all $n \geq N$, and since $a_n \leq M_n$, we see $a_n < M + \epsilon$ for all $n \geq N$.

Observation 2: For any $\epsilon > 0$, and any $n \geq 0$, $M_n \geq M > M - \epsilon$, because $\{M_n\}$ is nonincreasing and M is finite. Since $M_n = \sup\{a_k, k \geq n\}$, there is some $k \geq n$ such that $a_k > M - \epsilon$.

Now for $i = 1$, take n_1' such that $a_n < M + 1$ for all $n \geq n_1' + 1$. Property 2 implies that there is $n_1 \geq n_1' + 1$ such that $a_{n_1} > M - 1$. So $|a_{n_1} - M| < 1$.

Next, there exists $n_2' \geq n_1$ such that $a_n < M + \frac{1}{2}$ for all $n \geq n_2' + 1$. Property 2 implies that there is $n_2 \geq n_2' + 1 > n_1$ such that $a_{n_2} > M - \frac{1}{2}$. So $|a_{n_2} - M| < \frac{1}{2}$. Note that $n_2 > n_1$.

An inductive argument shows that one can construct $n_1 < n_2 < n_3 < \dots$ such that $|a_{n_k} - M| < \frac{1}{k}$ for all $k \geq 1$. This implies that $\lim_{k\to\infty} a_{n_k} = M$.

Step 3. Finally we show that $M = \limsup_{n\to\infty} a_n$. Consider some subsequence $\{a_{n_k}\}$ of $\{a_n\}$. Fix $\epsilon > 0$, and consider the integer N from Observation 1. Then $n_k \geq N$ for all $k \geq N$, so that $a_{n_k} < M + \epsilon$ for all $k \geq N$. This implies that $\{a_{n_k}\}$ does not diverge to ∞. Moreover, if $\{a_{n_k}\}$ converges, then $\lim_{k\to\infty} a_{n_k} \leq M + \epsilon$, and since $\epsilon > 0$ was arbitrarily fixed, this shows that $\lim_{k\to\infty} a_{n_k} \leq M$. So $M = \limsup_{n\to\infty} a_n$. □

From the above proof, we see the following:

- Suppose $\{a_n\}$ is bounded above. Then

$$\limsup_{n\to\infty} a_n = \lim_{n\to\infty} \sup_{k\geq n} a_k = \inf_{n\geq 1} \sup_{k\geq n} a_k.$$

  In this case, this value may be $-\infty$ or finite, but not ∞. If this value is $-\infty$, then $\lim_{n\to\infty} a_n = -\infty$.

- Similarly, if $\{a_n\}$ is bounded below, then

$$\liminf_{n\to\infty} a_n = \lim_{n\to\infty} \inf_{k\geq n} a_k = \sup_{n\geq 1} \inf_{k\geq n} a_k.$$

  If this value is ∞, then $\lim_{n\to\infty} a_n = \infty$.

- $\limsup_{n\to\infty} a_n = \infty$ if and only if $\{a_n\}$ is unbounded from above.

- $\liminf_{n\to\infty} a_n = -\infty$ if and only if $\{a_n\}$ is unbounded from below.

From the definitions, $\liminf_{n\to\infty} a_n \le \limsup_{n\to\infty}$. The next statement is about when equality takes place. The above observations are useful in its proof.

Theorem 22 $\liminf_{n\to\infty} a_n = \limsup_{n\to\infty} a_n$ *if and only if* $\lim_{n\to\infty} a_n$ *exists; in that case,* $\lim_{n\to\infty} a_n = \liminf_{n\to\infty} a_n = \limsup_{n\to\infty} a_n$.

Proof. The "if" part was known. We prove the "only if" part as follows. Case 1. Suppose $\liminf_{n\to\infty} a_n = \limsup_{n\to\infty} a_n = \infty$. Then, according to the observations made above this statement, $\{a_n\}$ is bounded from below, and $\lim_{n\to\infty} a_n = \infty$, as claimed.

Case 2. Suppose $\liminf_{n\to\infty} a_n = \limsup_{n\to\infty} a_n = -\infty$. Then, according to the observations made above this statement, $\{a_n\}$ is bounded from above, and $\lim_{n\to\infty} a_n = -\infty$, as claimed.

Case 3. Suppose $\liminf_{n\to\infty} a_n = \limsup_{n\to\infty} = M \in (-\infty, \infty)$. Then, according to the observations made above this statement, $\lim_{n\to\infty} \sup_{k\ge n} a_k = \lim_{n\to\infty} \inf_{k\ge n} a_k = M$. Let $\epsilon > 0$ be fixed. It follows that for some $N \ge 1$, $\sup_{k\ge n} a_k < M+\epsilon$ and $\inf_{k\ge n} a_k > M-\epsilon$ for all $n \ge N$. Hence, $M-\epsilon < \inf_{k\ge n} a_k \le a_n \le \sup_{k\ge n} a_k < M+\epsilon$ for all $n \ge N$. Thus, $\lim_{n\to\infty} a_n = M$. □

The next fact will be referred to later.

Lemma 5 *For any sequence of (strictly) positive numbers* $\{c_n\}$,

$$\liminf_{n\to\infty} \frac{c_{n+1}}{c_n} \le \liminf_{n\to\infty} (c_n)^{1/n} \le \limsup_{n\to\infty} (c_n)^{1/n} \le \limsup_{n\to\infty} \frac{c_{n+1}}{c_n}.$$

Chapter 3

Series

3.1 Convergent and divergent series

Let $\{a_k\}$ be a sequence of real numbers. Using this sequence, we can form a new sequence $\{s_k\}$ by putting $s_n = \sum_{k=1}^{n} a_k$. This chapter focuses on the convergence and divergence of the sequence $\{s_n\}$.

Introduce the notation $\sum_{k=1}^{\infty} a_k$ and call it a series. For each $k \in \mathbb{N}$, s_n is called a partial sum of this series.

Definition 20 (Convergent and divergent series) *We say the series $\sum_{k=1}^{\infty} a_k$ converges if $\{s_n\}$ converges. If a series does not converge, then it is called divergent. The series is said to diverge to ∞ or $-\infty$ if $\{s_n\}$ diverges to ∞ or $-\infty$, respectively. We write $\sum_{k=1}^{\infty} a_k = c$ if $\lim_{k\to\infty} s_n = c$ with $c \in [-\infty, \infty]$, and call c the value of the series.*

3.1.1 Elementary properties

The following assertions follow immediately from their counterparts for convergent sequences:

- If $\displaystyle\sum_{k=1}^{\infty} a_k$ converges, then its value is unique.
- If $\displaystyle\sum_{k=1}^{\infty} a_k$ and $\displaystyle\sum_{k=1}^{\infty} b_k$ converge, then so does the series $\sum_{k=1}^{\infty} \big(a_k + b_k\big)$, and it holds that
$$\sum_{k=1}^{\infty} \big(a_k + b_k\big) = \sum_{k=1}^{\infty} a_k + \sum_{k=1}^{\infty} b_k.$$
- If $\displaystyle\sum_{k=1}^{\infty} a_k$ converges, then so does the series $\displaystyle\sum_{k=1}^{\infty} ca_k$, and it holds that
$$\sum_{k=1}^{\infty} ca_k = c\sum_{k=1}^{\infty} a_k.$$
- If $a_k \geq 0$, then $\sum_{k=1}^{\infty} a_k$ either converges or diverges to ∞.
- If $\displaystyle\sum_{k=1}^{\infty} a_k$ converges, then $a_k \to 0$ as $k \to \infty$. Indeed, denoting $\sum_{k=1}^{\infty} a_k = \lim_{n\to\infty} s_n = C \in \mathbb{R}$, it holds that
$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} (s_n - s_{n-1}) = \lim_{n\to\infty} s_n - \lim_{n\to\infty} s_{n-1} = C - C = 0.$$

The last item implies that if $\{a_k\}$ does not converge to 0, then the series $\sum_{k=1}^{\infty} a_k$ does not converge. Sometimes, this is referred to as the null test. Note also that it can happen that $\{a_k\}$ converges to 0 but the series $\sum_{k=1}^{\infty} a_k$ does not converge.

Theorem 23 (Comparison test) *Consider two series* $\sum_{k=1}^{\infty} a_k$ *and* $\sum_{k=1}^{\infty} b_k$ *such that* $0 \le a_k \le b_k$ *for all* k.

(a) If $\sum_{k=1}^{\infty} b_k$ *converges, then so does* $\sum_{k=1}^{\infty} a_k$.

(b) If $\sum_{k=1}^{\infty} a_k$ *diverges, then so does* $\sum_{k=1}^{\infty} b_k$.

One can prove the above theorem easily by considering the corresponding sequences of partial sums of the two series, and by applying the monotone convergence theorem. The detail can be safely omitted here.

As a consequence of the previous result, we can show the following, which is also called a comparison test.

Theorem 24 (Comparison test') *If the terms of* $\sum_{k=1}^{\infty} a_k$ *and* $\sum_{k=1}^{\infty} b_k$ *are all positive, and if* $\lim_{k\to\infty} \frac{a_k}{b_k} = l \in (0,\infty)$, *then* $\sum_{k=1}^{\infty} a_k$ *converges if and only if* $\sum_{k=1}^{\infty} b_k$ *converges.*

Proof. Suppose $\sum_{k=1}^{\infty} b_k$ converges. Fix $\epsilon > 0$. From $\lim_{k\to\infty} \frac{a_k}{b_k} = l \in (0,\infty)$, we see that, for some natural number $N \ge 1$, $b_k(l-\epsilon) < a_k < b_k(l+\epsilon)$. Since $\sum_{k=1}^{\infty} b_k$ converges, so does $\sum_{k=N+1}^{\infty} b_k$. Thus, $(l+\epsilon)\sum_{k=N+1}^{\infty} b_k = \sum_{k=N+1}^{\infty} b_k(l+\epsilon)$ converges. Now by the previous theorem, we see that $\sum_{k=N+1}^{\infty} a_k$ converges. But this is the same as the convergence of $\sum_{k=1}^{\infty} a_k$.

The case of the convergence of $\sum_{k=1}^{\infty} a_k$ implying that of $\sum_{k=1}^{\infty} b_k$ can be seen by the similar argument. □

Example 32 *Show the convergence of the series* $\sum_{k=1}^{\infty} \frac{k+3}{k^3+2k^2+3k+1}$.

Solution. Note that

$$\frac{k+3}{k^3+2k^2+3k+1} = \frac{1+3/k}{k^2(1+2/k+3/k^2+1/k^3)} \le \frac{4}{k^2}.$$

Since the series $\sum_{k=1}^{\infty} \frac{4}{k^2}$ converges, and so does the series $\sum_{k=1}^{\infty} \frac{k+3}{k^3+2k^2+3k+1}$ by the comparison test. □

Example 33 *Show the divergence of the series* $\sum_{k=1}^{\infty} \sqrt{\frac{k+4}{k^2+3k+1}}$.

Solution. Note that

$$\sqrt{\frac{k+4}{k^2+3k+1}} = \sqrt{\frac{1+4/k}{k(1+3/k+1/k^2)}} \ge \sqrt{\frac{1}{5k}}.$$

Since the series $\sum_{k=1}^{\infty} \frac{1}{\sqrt{k}}$ diverges, and so does the series $\sum_{k=1}^{\infty} \sqrt{\frac{k+4}{k^2+3k+1}}$ by the comparison test. □

Consider the sequence $\{a_k\}$. Let M be a natural number. If we ignore the first M terms in $\{a_k\}$, we can form a new sequence $\{a_{M+k}\}_{k\ge1}$ or $\{a_k\}_{k>M}$. The series of the sequence $\{a_{M+k}\}_{k\ge1}$ or $\{a_k\}_{k>M}$ is denoted by $\sum_{k=1}^{\infty} a_{M+k}$ or $\sum_{k=M+1}^{\infty} a_k$. Then the series $\sum_{k=1}^{\infty} a_{M+k}$ converges if $\{\sum_{k=1}^{n} a_{M+k}\}$ converges, i.e., if $\{\sum_{k=M+1}^{M+n} a_k\}$ converges. Otherwise, that series diverges.

Intuitively speaking, the series $\sum_{k=1}^{\infty} a_{M+k}$ is obtained by ignoring the first M summands in the series $\sum_{k=1}^{\infty} a_k$. The next result asserts that $\sum_{k=1}^{\infty} a_{M+k}$ converges if and only if $\sum_{k=1}^{\infty} a_k$ converges.

Theorem 25 *Let* $M \ge 1$ *be a natural number. Then the series* $\sum_{k=1}^{\infty} a_k$ *converges if and only if the series* $\sum_{k=1}^{\infty} a_{M+k}$ *(i.e., the series* $\sum_{k=M+1}^{\infty} a_k$*) converges.*

Proof. Let $s_n = \sum_{k=1}^{n} a_k$. Then $s_n \to l$ for some real number l if and only if s_{n+M} converges to l, too, as $n \to \infty$. Write (for $n \in \mathbb{N}$) s_{n+M} out as

$$\begin{aligned} s_{n+M} &= \sum_{k=1}^{n+M} a_k = \sum_{k=1}^{M} a_k + \sum_{k=M+1}^{n+M} a_k \\ &= \sum_{k=1}^{M} a_k + \sum_{j=1}^{n} a_{M+j}, \end{aligned}$$

so that

$$s_{n+M} - \sum_{k=1}^{M} a_k = \sum_{j=1}^{n} a_{M+j}.$$

Observe that the left hand side converges as $n \to \infty$ if and only if so does the right hand side. This means, $\sum_{k=1}^{\infty} a_k$ converges if and only if $\sum_{j=1}^{\infty} a_{M+j}$ converges. □

3.1.2 Examples

Example 34 *Show the following series converges, and find its value:*

$$\sum_{k=1}^{\infty} \frac{a}{k(k+n_0)}$$

with $n_0 \in \mathbb{N}$ and $a \in \mathbb{R}$.

Solution. Write for $m > n_0$

$$\begin{aligned} n_0 \sum_{k=1}^{m} \frac{1}{k(k+n_0)} &= \sum_{k=1}^{m} \left(\frac{1}{k} - \frac{1}{k+n_0} \right) = \sum_{k=1}^{n_0} \frac{1}{k} + \sum_{k=n_0+1}^{m} \frac{1}{k} - \sum_{k=1}^{m} \frac{1}{k+n_0} \\ &= \sum_{k=1}^{n_0} \frac{1}{k} + \sum_{k=n_0+1}^{m} \frac{1}{k} - \sum_{j=n_0+1}^{m+n_0} \frac{1}{j} \\ &= \sum_{k=1}^{n_0} \frac{1}{k} - \sum_{j=m+1}^{m+n_0} \frac{1}{j}. \end{aligned}$$

Note that

$$0 \le \sum_{j=m+1}^{m+n_0} \frac{1}{j} \le \sum_{j=m+1}^{m+n_0} \frac{1}{m} = \frac{n_0}{m}$$

and $\frac{n_0}{m} \to 0$ as $m \to \infty$. By the squeeze theorem, $\sum_{j=m+1}^{m+n_0} \frac{1}{j} \to 0$ as $m \to \infty$. Therefore, $n_0 \sum_{k=1}^{m} \frac{1}{k(k+n_0)} \to \sum_{k=1}^{n_0} \frac{1}{k}$ as $m \to \infty$. This implies that

$$\sum_{k=1}^{\infty} \frac{a}{k(k+n_0)} = \frac{a}{n_0} \sum_{k=1}^{n_0} \frac{1}{k}.$$

Example 35 (Geometric series) *The series $\sum_{k=1}^{\infty} r^{k-1}$ is called the geometric series. From the formula*

$$\sum_{k=1}^{n} r^{k-1} = \frac{1-r^n}{1-r} \quad (r \neq 1),$$

we see that the geometric series converges if $|r| < 1$ with

$$\sum_{k=1}^{\infty} r^{k-1} = \lim_{n\to\infty} \frac{1-r^n}{1-r} = \frac{1}{1-r}.$$

The geometric series diverges if $|r| \ge 1$.

Example 36 (Harmonic series) *The series $\sum_{k=1}^{\infty} \frac{1}{k}$ is called the harmonic series. We have seen that the nondecreasing sequence of its partial sums $\{\sum_{k=1}^{n} \frac{1}{k}\}$ is not bounded above, and thus does not converge. In fact, $\sum_{k=1}^{\infty} \frac{1}{k} = \infty$. Note that the sequence $\{\frac{1}{k}\}$ converges to 0, but the harmonic series diverges.*

Example 37 (Alternating harmonic series) *The series*

$$\sum_{k=1}^{\infty}(-1)^{k-1}\frac{1}{k} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots,$$

is called the alternating harmonic series. Show that the alternating harmonic series converges.

Proof. By denoting the partial sums by

$$s_n = \sum_{k=1}^{n}(-1)^{k-1}\frac{1}{k},$$

we see that the subsequence $\{s_{2n}\}$ is increasing and $\{s_{2n-1}\}$ is decreasing. Furthermore, we have

$$s_2 = \frac{1}{2} \le s_{2n} \le s_{2n-1} \le 1 = s_1.$$

By the monotonic convergence theorem, we have that both limits

$$\lim_{n\to\infty} s_{2n} \quad \text{and} \quad \lim_{n\to\infty} s_{2n-1}$$

exist. Finally, since

$$s_{2n} - s_{2n-1} = -\frac{1}{2n} \to 0$$

we can conclude that $\{s_n\}$ converges. □

3.2 A kind of associative law

The series $\sum_{k=1}^{\infty} a_k$ can be understood as an ordered sum of infinitely many terms. For the sum of finitely many terms, i.e., a finite sum, one can apply the associative law: adding brackets does not change the value of the sum. The next result gives a version of the associative law for convergent series and series with nonnegative terms.

More precisely, given the series $\sum_{k=1}^{\infty} a_k$, one can form a new series $\sum_{k=1}^{\infty} b_k$ by grouping, i.e., $b_1 = (a_1 + \cdots + a_{n_1})$, $b_2 = (a_{n_1+1} + a_{n_1+2} + \cdots + a_{a_{n_2}})$, …, where $\{n_k\}$ is any increasing sequence of natural numbers. We say the associative law applies to the series $\sum_{k=1}^{\infty} a_k$, if $\sum_{k=1}^{\infty} b_k$ and $\sum_{k=1}^{\infty} a_k$ have the same value.

Theorem 26 (A kind of associative law for series) *Consider the series $\sum_{k=1}^{\infty} a_k$.*

(a) If $a_k \ge 0$ for all $k \in \mathbb{N}$, then the associative law applies to $\sum_{k=1}^{\infty} a_k$.

(b) If $\sum_{k=1}^{\infty} a_k$ converges, then the associative law applies to $\sum_{k=1}^{\infty} a_k$.

Proof. (a) Let $s_n = \sum_{k=1}^{n} a_k$ and $S_i = \sum_{k=1}^{i} b_k$. Note that (S_i) is a subsequence of (s_n), because for each natural number $i \ge 1$, $S_i = \sum_{k=1}^{n_i} a_k$, and $\{n_i\}$ is an increasing sequence of natural numbers. Therefore, if $\lim_{n\to\infty} s_n = l \in [0,\infty]$, then $\lim_{i\to\infty} S_i = l$.

On the other hand, for the nondecreasing sequence $\{s_n\}$, if $\lim_{i\to\infty} S_i = l \in [0,\infty]$, then $\lim_{n\to\infty} s_n = l$, completing the proof of (a). Note that $l = \sup\{S_i, i \in \mathbb{N}\}$.

We verify the above assertion in case $0 \le l < \infty$. The case of $l = \infty$ can be verified in a similar way. From $l = \sup\{S_i, i \in \mathbb{N}\}$, we see that $l \ge s_k$ for all $i \in \mathbb{N}$, because for all $i \in \mathbb{N}$, $n_i \ge i$ so that $S_i \ge s_i$.

Now for any $\epsilon > 0$, there is some $I \in \mathbb{N}$ such that $i \ge I$ implies $|S_i - l| = |\sum_{k=1}^{n_i} a_k - l| = l - \sum_{k=1}^{n_i} a_k < \epsilon$. In particular, $|\sum_{k=1}^{n_I} a_k - l| = l - \sum_{k=1}^{n_I} a_k < \epsilon$. Now let $N = n_I$. We have for all $n \ge N$ that $0 \le l - \sum_{k=1}^{n} a_k \le l - \sum_{k=1}^{n_I} < \epsilon$.

(b) The first paragraph in the proof of part (a) implies the assertion in (b). □

In general, the associative law does not hold for series. The sequence obtained by grouping can converge, while the original sequence does not.

Example 38 (p-Harmonic series) *For any parameter $0 < p < \infty$, the series*

$$\sum_{k=1}^{\infty}\frac{1}{k^p} = 1 + \frac{1}{2^p} + \frac{1}{3^p} + \frac{1}{4^p} + \cdots,$$

is called the p-harmonic series. The 1-harmonic series is simply the harmonic series. Show that the p-harmonic series converges if $p > 1$ and diverges if $p \le 1$.

Proof. First we observe that if $0 < p < 1$, then

$$\frac{1}{k} < \frac{1}{k^p}.$$

Since the harmonic series has unbounded partial sums, so does this p-harmonic series.

Now consider $p > 1$. We consider the series $\sum_{k=1}^{\infty} b_k$ obtained after grouping the p-harmonic series with $n_k = 2^k - 1$, $b_1 = \sum_{i=1}^{n_1} \frac{1}{i^p} = 1$ and $b_{k+1} = \sum_{i=n_k+1}^{n_{k+1}} \frac{1}{i^p}$.

That is, $\sum_{k=1}^{\infty} b_k$ is in the form

$$\begin{aligned} 1 + \left\{ \frac{1}{2^p} + \frac{1}{3^p} \right\} + \left\{ \frac{1}{4^p} + \frac{1}{5^p} + \frac{1}{6^p} + \frac{1}{7^p} \right\} + \left\{ \frac{1}{8^p} + \cdots + \frac{1}{15^p} \right\} + \cdots \\ \leq 1 + \frac{2}{2^p} + \frac{4}{4^p} + \frac{8}{8^p} + \cdots \\ \leq \frac{1}{1 - 2^{1-p}} \end{aligned}$$

Now

$$\begin{aligned} & b_1 = 1, \\ & b_{k+1} \leq \sum_{i=n_k+1}^{n_{k+1}} \frac{1}{(n_k+1)^p} = \sum_{i=n_k+1}^{n_{k+1}} \frac{1}{2^{kp}} = \frac{n_{k+1} - n_k}{(2^k)^p} = \frac{2^{k+1} - 2^k}{(2^k)^p} = \frac{2^k}{(2^k)^p}. \end{aligned}$$

Thus, $0 \leq \sum_{k=1}^{n} b_k \leq \sum_{k=1}^{n} \frac{2^k}{(2^k)^p} = \sum_{k=1}^{n} \left(\frac{1}{2^{p-1}}\right)^k \leq \frac{1}{1-2^{1-p}}$ for all $n \in \mathbb{N}$.

It follows from the monotone convergence theorem that the series $\sum_{k=1}^{\infty} b_k$ converges, and so does the p-harmonic series with $p > 1$, by Theorem 26. □

3.3 Absolutely convergent series

Theorem 27 *If the series* $\sum_{k=1}^{\infty} |a_k|$ *converges, then so does the series* $\sum_{k=1}^{\infty} a_k$.

Proof. We prove it by using the Bolzano-Cauchy convergence criterion. Let $\epsilon > 0$ be fixed. The convergence of $\sum_{k=1}^{\infty} |a_k|$ implies that for some natural number $N \geq 1$, $|\sum_{k=1}^{n} |a_k| - \sum_{k=1}^{m} |a_k|| < \epsilon$ for all $m, n \geq N$.

For $m, n \geq N$, if $m > n$, then

$$|\sum_{k=1}^{n} a_k - \sum_{k=1}^{m} a_k| = |\sum_{k=n+1}^{m} a_k| \leq \sum_{k=n+1}^{m} |a_k| = |\sum_{k=1}^{n} |a_k| - \sum_{k=1}^{m} |a_k|| < \epsilon.$$

That $|\sum_{k=1}^{n} a_k - \sum_{k=1}^{m} a_k| < \epsilon$ holds for $n > m$ and $n = m$, too. Therefore, $(\sum_{k=1}^{n} a_k)$ is Cauchy, and it converges by the Bolzano-Cauchy convergence criterion. □

Note that the above is only a sufficient condition for convergence of a series.

Definition 21 (Absolutely convergent series) *A series* $\sum_{k=1}^{\infty} a_k$ *is said to be absolutely convergent if the series* $\sum_{k=1}^{\infty} |a_k|$ *converges.*

For a real number a, let $a^+ = \max\{0, a\}$ and $a^- = \max\{0, -a\}$. They are called the positive and negative parts of a. Then $|a| = a^+ + a^-$ and $a = a^+ - a^-$. Now if $\sum_{n=1}^{\infty} a_n$ absolutely converges, then $\sum_{n=1}^{\infty} a_n^+$ and $\sum_{n=1}^{\infty} a_n^-$ both converge. On the other hand, if $\sum_{n=1}^{\infty} a_n^+$ and $\sum_{n=1}^{\infty} a_n^-$ are both convergent, then $\sum_{n=1}^{\infty} a_n^+ + \sum_{n=1}^{\infty} a_n^+ = \sum_{n=1}^{\infty} |a_n|$ converges. Hence, $\sum_{n=1}^{\infty} a_n$ absolutely converges if and only if $\sum_{n=1}^{\infty} a_n^+$ and $\sum_{n=1}^{\infty} a_n^-$ both converge.

Definition 22 (Conditionally convergent series) *A series* $\sum_{k=1}^{\infty} a_k$ *is said to be conditionally convergent if it converges but the series* $\sum_{k=1}^{\infty} |a_k|$ *diverges.*

Example 39 *The alternating harmonic series* $\sum_{k=1}^{\infty}(-1)^{k-1}\frac{1}{k}$ *is conditionally convergent.*

3.4 Rearrangement of series

A rearrangement of a series is defined as follows. Let $\sigma : \mathbb{N} \to \mathbb{N}$ be any permutation of the natural numbers. That is, σ is a bijection from $\mathbb{N}$ onto itself. If $b_n := a_{\sigma(n)}$ then $\{b_n\}$ is said to be a *rearrangement* of $\{a_n\}$, and $\sum_{n=1}^{\infty} b_n$ is called a rearrangement of $\sum_{n=1}^{\infty} a_n$.

Theorem 28 (Dirichlet) *Let* $\sum_{n=1}^{\infty} a_n$ *be an absolutely convergent series of real numbers. If* $\{b_n\}$ *is a rearrangement of* $\{a_n\}$*, then the following assertions hold.*

(a) $\sum_{n=1}^{\infty} b_n$ *is an absolutely convergent series.*

(b) $\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} a_n$.

Proof. Step 1. Consider the case $a_n \geq 0$ for all $n \geq 1$. Recall that $b_n = a_{\sigma(n)}$ with σ from $\mathbb{N}$ to $\mathbb{N}$ being one-one and onto.

For $M \geq 1$, consider the partial sum $\sum_{n=1}^{M} b_n$. There is some $M' \geq 1$ such that $\sigma(n) \leq M'$ for all $n \in \{1, \ldots, M\}$. It follows that $\sum_{n=1}^{M} b_n \leq \sum_{n=1}^{M'} a_n$. Since $\sum_{n=1}^{\infty} a_n$ absolutely converges, we see that $\sum_{n=1}^{M} b_n \leq \sum_{n=1}^{\infty} a_n \in [0, \infty)$. By the monotone convergence theorem, we see that $\sum_{n=1}^{M} b_n$ absolutely converges as $M \to \infty$, and $\sum_{n=1}^{\infty} b_n \leq \sum_{n=1}^{\infty} a_n$.

For the opposite direction of the previous inequality, note from $b_n = a_{\sigma(n)}$ that $a_n = b_{\sigma^{-1}(n)}$. So replace in the argument in the previous paragraph a with b, b with a and σ with σ^{-1}, we see $\sum_{n=1}^{\infty} a_n \leq \sum_{n=1}^{\infty} b_n$. Therefore, $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} b_n$.

Step 2. Now consider the general case when a_n may not be all positive. Let us prove that $\sum_{n=1}^{\infty} b_n$ absolutely converges. This follows from applying the result proved in Step 1 to $\sum_{n=1}^{\infty} |b_n|$.

Step 3. Finally, we prove that in the general case, $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} b_n$. Since $\sum_{n=1}^{\infty} b_n$ is absolutely convergent, as shown in Step 2, we may write $\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} b_n^+ - \sum_{n=1}^{\infty} b_n^-$, where the two series on the right hand side are both absolutely convergent. Likewise, we may write $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} a_n^+ - \sum_{n=1}^{\infty} a_n^-$. Observe that $b_n^+ = a_\sigma(n)^+$ and $b_n^- = a_\sigma(n)^-$. Now applying the result proved in Step 1 to $\sum_{n=1}^{\infty} b_n^+$ and $\sum_{n=1}^{\infty} b_n^-$, we see that $\sum_{n=1}^{\infty} b_n^+ = \sum_{n=1}^{\infty} a_n^+$ and $\sum_{n=1}^{\infty} b_n^- = \sum_{n=1}^{\infty} a_n^-$. Therefore, $\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} b_n^+ - \sum_{n=1}^{\infty} b_n^- = \sum_{n=1}^{\infty} a_n^+ - \sum_{n=1}^{\infty} a_n^- = \sum_{n=1}^{\infty} a_n$. □

The next example demonstrates that the requirement of absolute convergence is important in the above Dirichlet theorem.

Example 40 *Consider the alternating harmonic series* $\sum_{k=1}^{\infty}(-1)^{k-1}\frac{1}{k} = \sum_{k=1}^{\infty} a_k$ *with* $a_k = (-1)^{k-1}\frac{1}{k}$*. We know that it is conditionally convergent, so that we may put* $\sum_{k=1}^{\infty}(-1)^{k-1}\frac{1}{k} = A$ *for some* $A \in \mathbb{R}$*. We have seen from the proof of the convergence of alternating harmonic series that* $A \geq \frac{1}{2}$*. Let* $\gamma = \frac{A}{2}$*. Let us show that there is a rearrangement of this alternating harmonic series, say* $\sum_{k=1}^{\infty} b_k$ *with* $b_k = a_{\sigma(k)}$ *for the rearrangement* σ *such that* $\sum_{k=1}^{\infty} b_k = \frac{A}{2} \neq A$*.*

Solution. Let $\sigma(1) = 1$, and for each natural number $k \geq 1$,

$$\begin{aligned}
\sigma(3k-1) &= 4k-2 = 2(2k-1),\\
\sigma(3k) &= 4k = 2(2k),\\
\sigma(3k+1) &= 2k+1.
\end{aligned}$$

One can check that σ is indeed a bijection between two copies of the set of natural numbers.

Then the first few terms of (b_k) with $b_k = b_{\sigma(k)}$ read

$$1, -\frac{1}{2}, -\frac{1}{4}, \frac{1}{3}, -\frac{1}{6}, -\frac{1}{8}, \frac{1}{5}, -\frac{1}{10}, -\frac{1}{12}, \ldots, \frac{1}{2k-1}, -\frac{1}{4k-2}, -\frac{1}{4k}, \ldots.$$

Here $\frac{1}{2k-1} = b_{3k-2}$, $-\frac{1}{4k-2} = b_{3k-1}$, and $-\frac{1}{4k} = b_{3k}$.

Now consider the partial sum

$$\sum_{n=1}^{3k} b_n = \sum_{n=1}^{k}(\frac{1}{2n-1} - \frac{1}{4n-2} - \frac{1}{4n}) = \sum_{n=1}^{k}(\frac{1}{4n-2} - \frac{1}{4n}) = \frac{1}{2}\sum_{n=1}^{k}(\frac{1}{2n-1} - \frac{1}{2n}) = \frac{1}{2}\sum_{n=1}^{2k}(-1)^{n-1}\frac{1}{n}.$$

Hence, $\sum_{n=1}^{3k} b_n = s_{3k}$ converges to $\frac{A}{2}$.

Let us argue that (s_k) converges to $\frac{A}{2}$ as follows. Let $\epsilon > 0$ be fixed. Then there is some natural number $N \geq 1$ such that $|s_{3k} - \frac{A}{2}| < \frac{\epsilon}{4}$ for all $k \geq N$.

Note that $s_{3k+1} = \frac{1}{2k+1} + s_{3k}$ and $s_{3k+2} = s_{3k+1} - \frac{1}{4k+2}$. Let $M_1 \geq N$ be such that $\frac{1}{2k+1} < \frac{\epsilon}{4}$ so that $|s_{3k+1} - \frac{A}{2}| < \frac{\epsilon}{2}$ for all $k \geq M_1$. Also let $M_2 \geq M_1$ be such that $\frac{1}{4k+2} < \frac{\epsilon}{2}$ for all $k \geq M_2$. Then $|s_{3k+2} - \frac{A}{2}| < \epsilon$ for all $k \geq M_2$.

Finally, for all $k \geq 3M_2 + 2$, $|s_k - \frac{A}{2}| < \epsilon$. Since $\epsilon > 0$ was fixed arbitrarily, this shows the convergence of s_k to $\frac{A}{2}$ as $k \to \infty$, as required. □

The next result is called the Riemann theorem, after the German mathematician Riemann.

Theorem 29 (Riemann) *If the series* $\sum_{n=1}^{\infty} a_n$ *is conditionally convergent) and* $\gamma \in [-\infty, \infty]$ *is* any real number*, then there exists a rearrangement* $\{b_n\}$ *of the sequence* $\{a_n\}$ *so that*

$$\sum_{n=1}^{\infty} b_n = \gamma.$$

3.5 Product of series

The product of two finite sums $\sum_{n=1}^{M} a_n$ and $\sum_{n=1}^{M} b_n$ can be done as follows: work out the product of each term in $\sum_{n=1}^{M} a_n$ and each term in $\sum_{n=1}^{M} b_n$: $a_1b_1, \ldots, a_1b_M, a_2b_1, \ldots, a_2b_M, \ldots, a_Mb_1, \ldots, a_Mb_M$, and then take the sum of all of them. The result is $\sum_{n=1}^{M} a_n \sum_{n=1}^{M} b_n$.

Now one wishes to extend this rule to the two convergent series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$. Multiplying again each pair of the terms in the two series gives:

$$\begin{array}{cccccc} a_1b_1 & a_1b_2 & a_1b_3 & a_1b_4 & \cdots \\ a_2b_1 & a_2b_2 & a_2b_3 & a_2b_4 & \cdots \\ a_3b_1 & a_3b_2 & a_3b_3 & a_3b_4 & \cdots \\ a_4b_1 & a_4b_2 & a_4b_3 & a_4b_4 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{array}$$

Again, one may want to take the sum of all the above terms (more precisely series), and hopes that the series will be equal to $(\sum_{n=1}^{\infty} a_n)(\sum_{n=1}^{\infty} b_n)$.

The difficulty is the following: there are different orders according to which the terms in the above infinite array are added, and we have seen from the Riemann theorem that convergent series is in general sensitive to rearrangements.

More precisely, each order of taking the sums is given by a bijection $s \to (n_s, k_s)$ from $\mathbb{N}$ onto $\mathbb{N} \times \mathbb{N}$. The corresponding series will be $\sum_{s=1}^{\infty} a_{n_s} b_{k_s}$.

Theorem 30 (Cauchy) *Assume* $\sum_{n=1}^{\infty} a_n$, $\sum_{n=1}^{\infty} b_n$ *are absolutely convergent series of real numbers. Then for each* (n_s, k_s) *defined in the previous paragraph*

$$\sum_{s=1}^{\infty} a_{n_s} b_{k_s}$$

is absolutely convergent, and

$$\sum_{s=1}^{\infty} a_{n_s} b_{k_s} = (\sum_{n=1}^{\infty} a_n)(\sum_{n=1}^{\infty} b_n).$$

Proof. For each $M \geq 1$, consider the partial sum $\sum_{s=1}^{M} |a_{n_s} b_{k_s}|$. Let $J = \max\{n_s, k_s : \ s = 1, \ldots, M\}$. It follows that $\sum_{s=1}^{M} |a_{n_s} b_{k_s}| \leq (\sum_{j=1}^{J} |a_j|)(\sum_{j=1}^{J} |b_j|)$, because expanding the right hand side, we see that there are at least as many nonnegative terms being added up on the right as compared to the left. Since $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are absolutely convergent, by passing to the limit as $J \to \infty$, we see that $\sum_{s=1}^{M} |a_{n_s} b_{k_s}| \leq (\sum_{j=1}^{\infty} |a_j|)(\sum_{j=1}^{\infty} |b_j|) < \infty$. By the monotone convergence theorem, we see that $\sum_{s=1}^{M} |a_{n_s} b_{k_s}|$ converges, and the series $\sum_{s=1}^{\infty} a_{n_s} b_{k_s}$ converges absolutely.

The rest shows that $\sum_{s=1}^{\infty} a_{n_s} b_{k_s} = (\sum_{n=1}^{\infty} a_n)(\sum_{n=1}^{\infty} b_n)$.

Since $\sum_{s=1}^{\infty} a_{n_s} b_{k_s}$ converges absolutely, rearrangement of this series does not affect its value. Consider the following series obtained from a specific rearrangement:

$$\begin{aligned} & a_1 b_1 \\ & + a_1 b_2 + a_2 b_2 + a_2 b_1 \\ & + a_1 b_3 + a_2 b_3 + a_3 b_3 + a_3 b_2 + a_3 b_1 \\ & + \ldots \end{aligned}$$

By the Dirichlet theorem, this series is convergent (in fact, absolutely), so that we may apply the associative law to this series, and its value is the same as the one of the following series, whose terms correspond to the sums inside brackets:

$$\begin{aligned} & (a_1 b_1) \\ & + (a_1 b_2 + a_2 b_2 + a_2 b_1) \\ & + (a_1 b_3 + a_2 b_3 + a_3 b_3 + a_3 b_2 + a_3 b_1) \\ & + \ldots \end{aligned}$$

It remains to show that the value of this series is $(\sum_{n=1}^{\infty} a_n)(\sum_{n=1}^{\infty} b_n)$. Denote the partial sums of this series by S_m, and the partial sums of $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ by A_m and B_m, respectively. Then $S_1 = a_1 b_1 = A_1 B_1$, $S_2 = a_1 b_1 + a_1 b_2 + a_2 b_2 + a_2 b_1 = A_2 B_2$, $S_3 = a_1 b_1 + a_1 b_2 + a_2 b_2 + a_2 b_1 + a_1 b_3 + a_2 b_3 + a_3 b_3 + a_3 b_2 + a_3 b_1 = A_3 B_3$ and in general, $S_m = A_m B_m$. So we see that $S_m \to (\sum_{n=1}^{\infty} a_n)(\sum_{n=1}^{\infty} b_n)$ as $m \to \infty$, as required. □

Consider the following array:

$$\begin{array}{ccccc} a_1 b_1 & a_1 b_2 & a_1 b_3 & a_1 b_4 & \cdots \\ a_2 b_1 & a_2 b_2 & a_2 b_3 & a_2 b_4 & \cdots \\ a_3 b_1 & a_3 b_2 & a_3 b_3 & a_3 b_4 & \cdots \\ a_4 b_1 & a_4 b_2 & a_4 b_3 & a_4 b_4 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{array}$$

A special way of taking the sum of the following terms is along the diagonal, and grouping the terms along each diagonal:

$$\begin{aligned} & a_1 b_1 + (a_1 b_2 + a_2 b_1) + (a_1 b_3 + a_2 b_2 + a_3 b_1) + \ldots \\ = & \sum_{n=2}^{\infty} \sum_{k=1}^{n-1} a_k b_{n-k}. \end{aligned}$$

In view of the last theorem, if $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are absolutely convergent, then $\sum_{n=2}^{\infty} \sum_{k=1}^{n-1} a_k b_{n-k} = (\sum_{n=1}^{\infty} a_n)(\sum_{n=1}^{\infty} b_n)$. In fact the last equality holds under weaker conditions: if $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are convergent, and at least one of them is absolutely convergent, then $\sum_{n=2}^{\infty} \sum_{k=1}^{n-1} a_k b_{n-k} = (\sum_{n=1}^{\infty} a_n)(\sum_{n=1}^{\infty} b_n)$. This is known as the Mertens theorem. In fact, if $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ converge, then $\sum_{n=2}^{\infty} \sum_{k=1}^{n-1} a_k b_{n-k}$ cannot converge to anything other than $(\sum_{n=1}^{\infty} a_n)(\sum_{n=1}^{\infty} b_n)$. This fact is known as the Abel theorem.

It should be emphasized that the convergence of $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ does not imply the convergence of $\sum_{n=2}^{\infty} \sum_{k=1}^{n-1} a_k b_{n-k}$. This is demonstrated by the next example.

Example 41 *Let* $a_n = b_n = (-1)^{n-1}\frac{1}{\sqrt{n}}$ *for all* $n \geq 1$. *We will see in a subsequent section that* $\sum_{n=1}^{\infty} a_n$ *and* $\sum_{n=1}^{\infty} b_n$ *conditionally converge. Now for* $n \geq 2$, $\sum_{k=1}^{n-1} a_k b_{n-k} = \sum_{k=1}^{n-1}(-1)^{k-1}\frac{1}{\sqrt{k}}(-1)^{n-k-1}\frac{1}{\sqrt{n-k}} = (-1)^{n-2}\sum_{k=1}^{n-1}\frac{1}{\sqrt{k}\sqrt{n-k}}$ *so that* $|\sum_{k=1}^{n-1} a_k b_{n-k}| = \sum_{k=1}^{n-1}\frac{1}{\sqrt{k}\sqrt{n-k}} > \sum_{k=1}^{n-1}\frac{1}{\sqrt{n}\sqrt{n}} = 1$. *This implies that the series* $\sum_{n=2}^{\infty}\sum_{k=1}^{n-1} a_k b_{n-k}$ *does not converge.*

Definition 23 (Cauchy product) *The series* $\sum_{n=2}^{\infty}\sum_{k=1}^{n-1} a_k b_{n-k}$ *is called the Cauchy product of* $\sum_{n=1}^{\infty} a_n$ *and* $\sum_{n=1}^{\infty} b_n$.

3.6 Test for convergence

3.6.1 d'Alembert test and ratio test

The next result is after d'Alembert.

Theorem 31 (d'Alembert) *Consider the series* $\sum_{n=1}^{\infty} a_n$, *where* $a_n > 0$ *for all* $n \geq 1$.

(a) If for some $N \geq 1$ *and* $l \in (0,1)$, $\frac{a_{n+1}}{a_n} \leq l$ *for all* $n \geq N$, *then* $\sum_{n=1}^{\infty} a_n$ *converges.*

(b) If for some $N \geq 1$, $\frac{a_{n+1}}{a_n} \geq 1$ *for all* $n \geq N$, *then* $\sum_{n=1}^{\infty} a_n$ *diverges.*

Proof. (a) For some $N \geq 1$, we have $\frac{a_{n+1}}{a_n} \leq l$ for all $n \geq N$, so that $a_{N+k} \leq l^k a_N$. The statement follows from an application of the comparison test.

(b) For some $N \geq 1$, $a_{N+n} \geq a_N > 0$ for all $n \geq 1$. By the comparison test, $\sum_{n=1}^{\infty} a_n$ diverges. □

It is often convenient to apply the above test in the following form.

Theorem 32 (Ratio test) *Suppose the terms of the series* $\sum_{k=1}^{\infty} a_k$ *are all strictly positive and*

$$\lim_{k\to\infty}\frac{a_{k+1}}{a_k} = l.$$

(Here l *is allowed to be* ∞.*)*

(a) If $l < 1$, *then the series converges.*

(b) If $l > 1$, *then the series diverges.*

Proof. (a) If

$$\lim_{k\to\infty}\frac{a_{k+1}}{a_k} < 1$$

then there is a number $\alpha < 1$ so that there exists N large enough s.t.

$$\frac{a_{k+1}}{a_k} < \alpha$$

for $k \geq N$. Now apply the above test of d'Alembert.

(b) In this case, there is some $N \geq 1$ such that for all $n \geq N$, $\frac{a_{n+1}}{a_n} \geq 1$, and the result follows from the test of d'Alembert. □

If in the above ratio test, $l = 1$, then the test is inconclusive in general, and the series $\sum_{k=1}^{\infty} a_k$ may diverge or converge. (Check $\sum \frac{1}{k^2}$ and $\sum \frac{1}{k}$.)

Example 42 *Show that the series* $\sum_{k=1}^{\infty}\frac{(k!)^2}{(2k)!}$ *converges.*

Solution. Note that

$$\frac{a_{k+1}}{a_k} = \frac{((k+1)!)^2}{(2k+2)!}\frac{(2k)!}{(k!)^2} = \frac{(k+1)^2}{(2k+2)(2k+1)} \to \frac{1}{4}.$$

Consequently, $\sum_{k=1}^{\infty}\frac{(k!)^2}{(2k)!}$ converges. □

3.6.2 Cauchy test and root test

The next result is after Cauchy.

Theorem 33 (Cauchy) *Consider $\sum_{n=1}^{\infty} a_n$ with $a_n \geq 0$ for all $n \geq 1$.*

(a) If for some $N \geq 1$, it holds that $(a_n)^{1/n} \leq l < 1$ for all $n \geq N$, then $\sum_{n=1}^{\infty} a_n$ is convergent.

(b) If for some $N \geq 1$, $(a_n)^{1/n} \geq 1$ for all $n \geq N$, then $\sum_{n=1}^{\infty} a_n$ diverges.

Proof. (a) For all $n \geq N$, we see from $(a_n)^{1/n} \leq l < 1$ that $a_n \leq l^n$ with $l \in [0, 1)$. Since $\sum_{n=N}^{\infty} l^n$ converges, we see from the comparison test that $\sum_{n=N}^{\infty} a_n$ converges, and hence $\sum_{n=1}^{\infty} a_n$ converges.

(b) For all $n \geq N$, $(a_n)^{1/n} \geq 1$ implies that $a_n \geq 1$, and $\sum_{n=N}^{\infty} 1$ diverges. By the comparison test, we see that $\sum_{n=1}^{\infty} a_n$ diverges. □

It is often convenient to apply the above test of Cauchy in the following form.

Theorem 34 (Root test) *Suppose that the terms of the series $\sum_{k=1}^{\infty} a_k$ are all nonnegative and the following limit exists:*

$$\lim_{k\to\infty} \sqrt[k]{a_k} = l.$$

(Here l may be ∞, but it cannot be negative.)

(a) If $l < 1$, then the series converges.

(b) If $l > 1$, then the series diverges.

Proof. (a) From $\lim_{n\to\infty}(a_n)^{1/n} = l < 1$, we see that for some $N \geq 1$, $(a_n)^{1/n} < l + \frac{1-l}{2} < 1$ for all $n \geq N$. It remains to apply the previous result of Cauchy.

(b) In case $l < \infty$, like in the above, we have, for some $N \geq 1$, $(a_n)^{1/n} > l - \frac{l-1}{2} \geq 1$ for all $n \geq N$. In case $l = \infty$, we have, for some $N \geq 1$, $(a_n)^{1/n} > 1$ for all $n \geq N$, too. Now one can apply the previous result of Cauchy. □

If in the above statement of root test $l = 1$, then the root test is inconclusive, i.e., $\sum_{k=1}^{\infty} a_k$ may diverge or converge. (Check $\sum \frac{1}{k^2}$ and $\sum \frac{1}{k}$.)

Example 43 *Let $r > 0$. Decide for what values of r the series $\sum_{k=1}^{\infty} kr^k$ converges.*

Solution. Note that

$$\lim_{k\to\infty} \left(kr^k\right)^{1/k} = \lim_{k\to\infty} \sqrt[k]{k}r = r.$$

So the series converges for all $0 < r < 1$ and diverges for all $r > 1$.

When $r = 1$, we have $\sum_{k=1}^{\infty} kr^k = \sum_{k=1}^{\infty} k$, which is obviously divergent. □

3.6.3 Alternating series test

The following result of Leibniz is called the alternating series test.

Theorem 35 (Leibniz) *The series*

$$\sum_{k=1}^{\infty} (-1)^{k-1} a_k$$

converges if the sequence $\{a_k\}$ is nonincreasing and converges to 0.

Proof. Denoting the partial sums by

$$s_n = \sum_{k=1}^{n} (-1)^{k-1} a_k,$$

we see that the subsequence $\{s_{2n}\}$ is nondecreasing and $\{s_{2n-1}\}$ is nondecreasing. Furthermore, we have

$$a_1 - a_2 \leq s_{2n} \leq s_{2n-1} \leq a_1.$$

By the monotone convergence theorem, we have that both limits

$$\lim_{n\to\infty} s_{2n} \quad \text{and} \quad \lim_{n\to\infty} s_{2n-1}$$

exist. Finally, since

$$s_{2n} - s_{2n-1} = a_{2n} \to 0,$$

as $n \to \infty$. We conclude that $\lim_{n\to\infty} s_n$ exists. □

Example 44 (Alternating p-harmonic series) *The series*

$$\sum_{k=1}^{\infty} (-1)^{k-1} \frac{1}{k^p} = 1 - \frac{1}{2^p} + \frac{1}{3^p} - \frac{1}{4^p} + \cdots,$$

for any parameter $0 < p < \infty$ is called the alternating p-harmonic series. By the alternating series test, the alternating p-harmonic series is converges for all $0 < p < \infty$. Furthermore, it is conditionally convergent for $0 < p \leq 1$, absolutely convergent for $p > 1$.

3.6.4 A theorem of Cauchy

Theorem 36 (Cauchy) *Consider $\sum_{k=1}^{\infty} a_k$. Suppose $\{a_k\}$ is a nonincreasing sequence of nonnegative numbers. Then $\sum_{k=1}^{\infty} a_k$ converges if and only if $\sum_{k=1}^{\infty} 2^k a_{2^k}$ converges.*

Proof. "If" part. Assume that $\sum_{k=1}^{\infty} 2^k a_{2^k}$ converges. Then for each $n \in \mathbb{N}$, there is some $m \in \mathbb{N}$ satisfying $n < 2^m$, which implies that

$$\begin{aligned}
&\sum_{k=1}^{n} a_k \leq \sum_{k=1}^{2^{m+1}-1} a_k \\
= \;& a_1 + (a_2 + a_3) + (a_4 + a_5 + a_6 + a_7) + \cdots + (a_{2^m} + \cdots + a_{2^{m+1}-1}) \\
\leq \;& a_1 + 2a_2 + 4a_4 + \cdots + 2^m a_{2^m} = a_1 + \sum_{k=1}^{m} 2^k a_{2^k} \leq a_1 + \sum_{k=1}^{\infty} 2^k a_{2^k},
\end{aligned}$$

where $\sum_{k=1}^{\infty} 2^k a_{2^k}$ is finite by assumption. It follows that $\{\sum_{k=1}^{n} a_k\}$ is a nondecreasing and bounded above sequence. By monotone convergence theorem, this sequence converges, meaning that $\sum_{k=1}^{\infty} a_k$ converges.

"Only if" part. Assume that $\sum_{k=1}^{\infty} a_k$ converges. Then for each $m \in \mathbb{N}$, there is some $n \in \mathbb{N}$ satisfying $n > 2^m$, which implies

$$\begin{aligned}
&\sum_{k=1}^{n} a_k \geq \sum_{k=1}^{2^m} a_k \\
= \;& a_1 + a_2 + (a_3 + a_4) + (a_5 + a_6 + a_7 + a_8) + \cdots + (a_{2^{m-1}+1} + \cdots + a_{2^m}) \\
\geq \;& a_1 + a_2 + 2a_4 + 4a_8 + \cdots + 2^{m-1} a_{2^m} \\
\geq \;& a_1 + \frac{1}{2} \sum_{k=1}^{m} 2^k a_{2^k},
\end{aligned}$$

i.e., $\sum_{k=1}^{m} 2^k a_{2^k} \leq 2\sum_{k=1}^{n} a_k - 2a_1 \leq 2\sum_{k=1}^{\infty} a_k < \infty$, and thus $\{\sum_{k=1}^{m} 2^k a_{2^k}\}$ is a nondecreasing bounded above sequence. That is, $\sum_{k=1}^{\infty} 2^k a_{2^k}$ converges. □

Example 45 *Consider the series*

$$\sum_{k=2}^{\infty} \frac{1}{k(\ln k)^p}$$

for any parameter $-\infty < p < \infty$.

If $p \leq 0$, *then* $\frac{1}{k(\ln k)^p} \geq \frac{1}{k}$ *for* $k \geq 3$, *and by the comparison test, we see that* $\sum_{k=3}^{\infty} \frac{1}{k(\ln k)^p} \geq \sum_{k=3}^{\infty} \frac{1}{k}$ *diverges. Thus,* $\sum_{k=2}^{\infty} \frac{1}{k(\ln k)^p}$ *diverges for* $p \leq 0$.

If $p > 0$, *then from*

$$\begin{aligned} \sum_{k=2}^{\infty} 2^k \frac{1}{2^k(\ln(2^k))^p} &= \sum_{k=2}^{\infty} 2^k \frac{1}{2^k k^p (\ln 2)^p} \\ &= \frac{1}{(\ln 2)^p} \sum_{k=2}^{\infty} \frac{1}{k^p}, \end{aligned}$$

we see from the above theorem of Cauchy that $\sum_{k=2}^{\infty} \frac{1}{k(\ln k)^p}$ *converges if* $p > 1$ *and diverges if* $p \leq 1$. *So to sum up,* $\sum_{k=2}^{\infty} \frac{1}{k(\ln k)^p}$ *converges if* $p > 1$ *and diverges if* $-\infty < p \leq 1$.

Chapter 4

Power Series

For us a power series is a series of the form

$$\sum_{n=0}^{\infty} a_n x^n$$

where $\{a_n\}$ is a sequence of real numbers and $x \in \mathbb{R}$. We consider $\{a_n\}$ as fixed for this series, and x as a variable.

The series might converge for some values of x and not for others. Given a power series, we would like to establish which values of x lead to convergence, and which lead to divergence.

Define the subset of real numbers

$$S := \left\{ x \in \mathbb{R} : \sum_{n=0}^{\infty} a_n x^n \text{ converges} \right\}.$$

Note that $S \neq \emptyset$, because $0 \in S$. The set S is referred as the set of convergence of the series. More often, it is called the convergence interval of the power series. It will be clear why this terminology is preferred shortly.

Example 46 *For the power series* $\sum_{n=0}^{\infty} x^n$, $S = (-1, 1)$.

4.1 Radius of Convergence

Lemma 6 *Consider the power series* $\sum_{n=0}^{\infty} a_n x^n$. *If for some* $y \neq 0$, $\sum_{n=0}^{\infty} a_n y^n$ *converges, then for all* x *satisfying* $|x| < |y|$, $\sum_{n=0}^{\infty} a_n x^n$ *absolutely converges.*

Proof. Since $\sum_{n=0}^{\infty} a_n y^n$ converges, $a_n y_n \to 0$ as $n \to \infty$. Therefore, for some $M > 0$, $|a_n y^n| \leq M$ for all $n \geq 0$.

Consider x such that $|x| < |y|$. Then $|a_n||x^n| = |a_n||y^n|\frac{|x^n|}{|y^n|} = |a_n||y^n|\left(\frac{|x|}{|y|}\right)^n \leq M\left(\frac{|x|}{|y|}\right)^n$. Since the series $\sum_{n=0}^{\infty} M\left(\frac{|x|}{|y|}\right)^n$ converges, by the comparison test, $\sum_{n=0}^{\infty} |a_n||x^n|$ converges, too. The claimed result follows. □

Consider the following cases.

Case 1. If $\sum_{n=0}^{\infty} a_n x^n$ only converges when $x = 0$, then for all x satisfying $|x| > 0$, the power series diverges, and for all x satisfying $|x| < 0$, the power series converges (in fact, converges absolutely); this is trivial, because there is no x satisfying $|x| < 0$.

Case 2. Suppose $\sum_{n=0}^{\infty} a_n x$ converges when $x = y$ for some $y \neq 0$. Then let

$$E = \{|y| : \ y \neq 0, \ \sum_{n=0}^{\infty} a_n y^n \text{converges}\}.$$

Then $E \neq \emptyset$.

(a) If E is unbounded, then for any given x, there is some $|y| \in E$ such that $x < |y|$ and $-x < |y|$, so that $|x| < |y|$. Hence, by the previous lemma, $\sum_{n=0}^{\infty} a_n x^n$ converges (in fact absolutely converges) for all $x \in (-\infty, \infty)$.

(b) If E is bounded, then one can consider $\sup E \in (-\infty, \infty)$. If x satisfies $|x| < \sup E$, then, for some $|y| \in E$, $|x| < |y|$, and by the previous lemma $\sum_{n=0}^{\infty} a_n x^n$ converges (in fact absolutely converges). If $|x| > \sup E$, then $|x| \notin E$, and by the definition of E, $\sum_{n=0}^{\infty} a_n x^n$ diverges.

The above observation leads to the next theorem.

Theorem 37 *Given a power series* $\displaystyle\sum_{n=0}^{\infty} a_n x^n$*, there is always a (unique) constant* $R \in [0, \infty]$ *such that for all* x*, if* $|x| < R$*, then* $\sum_{n=0}^{\infty} a_n x^n$ *converges (in fact, absolutely converges), whereas if* $|x| > R$*,* $\sum_{n=0}^{\infty} a_n x^n$ *diverges.*

In Case 1, $R = 0$; in Case 2(a), $R = \infty$; in Case 2(b), $R \in (0, \infty)$.

We refer to R as the *Radius of Convergence* of the power series $\sum_{n=0}^{\infty} a_n x^n$.

Remark 5 (Convergence interval) *It follows from the above theorem that*

$$(-R, R) \subseteq S \subseteq [-R, R]$$

and the power series converges absolutely in $(-R, R)$*. In particular we see that* S *is always an* interval *(in the extended real line), and therefore, it is also called the convergence interval of the underlying power series.*

It can happen that the inclusions in the above remark hold strictly.

Example 47 (Everything can happen at boundary) *Consider the power series* $\sum_{n=1}^{\infty} \frac{x^n}{n}$*.*

- *When* $|x| > 1$ *the general term* $x^n/n \not\to 0$*, and so the series diverges for* $|x| > 1$ *by the null test. (In greater detail, when* $|x| > 1$*,* $x^{2n}/2n = \frac{1}{2}\frac{(x^2)^n}{n} \to \infty$ *because exponential growth beats polynomial growth, so that* $x^n/n \not\to 0$*.)*
- *When* $0 < |x| < 1$ *the series converges (absolutely) by the comparison test because* $|x^n/n| \leq |x|^n$ *for all* $n \in \mathbb{N}$*.*

It follows that $R = 1$*.*

- *For* $x = 1$ *we have the harmonic series* $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n}$*, which diverges.*
- *For* $x = -1$ *we have the alternating series* $\displaystyle\sum_{n=1}^{\infty} (-1)^n \frac{1}{n}$*, which conditionally converges.*

It follows that $S = [-1, 1)$ *for this particular power series.*

Let us reiterate: at x with $|x| = R$, provided that $0 < R < \infty$, it can happen that the power series absolutely converges, or conditionally converges, or diverges. One has to check the situations at the boundary points separately.

4.2 Cauchy-Hadamard theorem

The next result came from Cauchy initially and then the French mathematician Hadamard later, independently, and is called the Cauchy-Hadamard theorem.

Theorem 38 (Cauchy-Hadamard) *Consider a power series* $\sum_{n=0}^{\infty} a_n x^n$*. Then its radius of convergence is given by* $R = \frac{1}{\rho}$*, where* $\rho = \limsup_{n\to\infty}(|a_n|)^{1/n}$*. Here we put* $\frac{1}{0} = \infty$ *and* $\frac{1}{\infty} = 0$*.*

Proof. Case 1. Suppose $\rho = 0$. We need to show in this case that $R = \infty$, i.e., $\sum_{n=0}^{\infty} a_n x^n$ converges for all $x \in (-\infty, \infty)$. Let x be fixed. Since $0 \leq \liminf_{n\to\infty}(|a_n|)^{1/n} \leq \limsup_{n\to\infty}(|a_n|)^{1/n} = 0$, we see that $\lim_{n\to\infty}(|a_n|)^{1/n} = 0$. Now $\lim_{n\to\infty}(|a_n||x|^n)^{1/n} = |x| \lim_{n\to\infty}(|a_n|)^{1/n} = 0$. By the root test of Cauchy, $\sum_{n=0}^{\infty} |a_n x^n|$ converges, and thus, $\sum_{n=0}^{\infty} a_n x^n$ converges.

Case 2. Suppose $\rho = \infty$. In this case, there is a subsequence $\{|a_{n_i}|^{1/n_i})$ of $\{|a_n|^1/n\}$ such that $\lim_{i\to\infty} |a_{n_i}|^{1/n_i} = \infty$. Let $x \neq 0$ be fixed. Then for some $K \geq 0$, $|a_{n_i}|^{1/n_i} > \frac{1}{|x|}$ for all $i \geq K$. Thus, $|a_{n_i}||x|^{n_i} > 1$ for all $i \geq K$. Therefore, $\{|a_n||x|^n\}$ does not converge to 0 as $n \to \infty$. It follows that $\sum_{n=0}^{\infty} a_n x^n$ does not converge absolutely.

Case 3. Suppose $\rho \in (0, \infty)$. Let x be such that $|x| < \frac{1}{\rho}$. For some $\epsilon > 0$, we have $|x| < \frac{1}{\rho+\epsilon}$. (Why?) Since $\rho \in (0, \infty)$, $\rho = \lim_{n\to\infty} \sup_{k\geq n} |a_k|^{1/k}$, there is some $N \geq 1$ such that for all $n \geq N$, $\sup_{k\geq n} |a_k|^{1/k} < \rho + \epsilon/2$, and thus, $|a_n|^{1/n} < \rho + \epsilon/2$ for all $n \geq N$. Then $(|a_n||x|^n)^{1/n} = |a_n|^{1/n}|x| < \frac{\rho+\epsilon/2}{\rho+\epsilon} < 1$ for all $n \geq N$. By the test of Cauchy, we see that $\sum_{n=0}^{\infty} |a_n x^n|$ converges. Thus, $\sum_{n=0}^{\infty} a_n x^n$ absolutely converges for all x such that $|x| < \frac{1}{\rho}$.

Next, fix x such that $|x| > \frac{1}{\rho}$. Then for some $\epsilon > 0$, $|x| > \frac{1}{\rho-\epsilon} > 0$. There is a subsequence $\{|a_{n_i}|^{1/n_i}\}$ of $\{|a_n|^{1/n}\}$ and some $I \in \mathbb{N}$ such that $|a_{n_i}|^{1/n_i} > \rho - \epsilon$ for all $i \geq I$. Then $(|a_{n_i}||x|^{n_i})^{1/n_i} = |a_{n_i}|^{1/n_i}|x| > \frac{\rho-\epsilon}{\rho-\epsilon} = 1$ for all $i \geq I$. This implies that $|a_{n_i} x^{n_i}| > 1$ for all $i \geq I$, and hence, $\{|a_n x^n|\}$ does not converge to 0 as $n \to \infty$. Consequently, $\sum_{n=0}^{\infty} a_n x^n$ diverges if $|x| > \frac{1}{\rho}$. □

We now formulate two statements that immediately follow from the Cauchy-Hadamard theorem, but are convenient for practical applications.

Theorem 39 (Root test for power series) *Consider the power series $\sum_{n=0}^{\infty} a_n x^n$. Suppose that $|a_n|^{\frac{1}{n}} \to \rho$ as $n \to \infty$. Then $R = \frac{1}{\rho}$.*

Proof. This is an immediate consequence of the Cauchy-Hadamard theorem. □

Theorem 40 (Ratio test for power series) *Consider the power series $\sum_{n=0}^{\infty} a_n x^n$, where $a_n \neq 0$ for all $n \geq 0$. Suppose that $\frac{|a_{n+1}|}{|a_n|} \to \rho$ as $n \to \infty$. Then $R = \frac{1}{\rho}$.*

Proof. This is an immediate consequence of the Cauchy-Hadamard theorem, because the condition in the statement implies that $\lim_{n\to\infty} |a_n|^{1/n} = \rho$. □

Example 48 *Find the radius of convergence of the power series $\sum_{n=0}^{\infty}(-1)^n n^2 x^n$. What is the convergence interval S for this power series?*

Solution. We shall use the ratio test. Let $a_n = (-1)^n n^2$ and observe that

$$\frac{|a_{n+1}|}{|a_n|} = \frac{(n+1)^2}{n^2} = \left(1 + \frac{1}{n}\right)^2 \to 1$$

as $n \to \infty$, by the algebra of limits. Hence the radius of convergence

$$R = \frac{1}{1} = 1.$$

Thus $(-1, 1) \subseteq S \subseteq [-1, 1]$, and so it remains to determine whether the power series converges for $x = \pm 1$. If $x = 1$, then the power series becomes the series

$$\sum_{n=1}^{\infty}(-1)^n n^2,$$

which diverges by the null test as $(-1)^n n^2 \not\to 0$. Thus $1 \notin S$. Similarly $-1 \notin S$, and so $S = (-1, 1)$. □

Example 49 *Find the radius of convergence of the power series $\sum_{n=0}^{\infty} \frac{x^n}{n!}$. What is the convergence interval S for this power series?*

Solution. Let $a_n = 1/n!$ and observe that

$$\frac{|a_{n+1}|}{|a_n|} = \frac{n!}{(n+1)!} = \frac{1}{n+1} \to 0$$

as $n \to \infty$. Hence $R = \infty$ by the ratio test for power series, and so the power series converges (absolutely) for all $x \in \mathbb{R}$; i.e. $S = \mathbb{R}$. □

4.3 Power series about a point

We call the series

$$\sum_{n=0}^{\infty} a_n(x-c)^n = a_0 + a_1(x-c) + a_2(x-c)^2 + a_3(x-c)^3 + \cdots,$$

a power series about the point c or centered at the point c, where $c \in \mathbb{R}$ is a constant.

If $\sum_{n=0}^{\infty} a_n x^n$ has the radius of convergence R, then

(a) $\sum_{n=0}^{\infty} a_n(x-c)^n$ (absolutely) converges when $|x-c| < R$; and

(b) $\sum_{n=0}^{\infty} a_n(x-c)^n$ diverges when $|x-c| > R$.

In this case, we shall still refer R to as the radius of convergence of the power series $\sum_{n=0}^{\infty} a_n(x-c)^n$ centered at $c \in \mathbb{R}$.

4.3.1 Term by term addition of power series

Let $\sum_{n=0}^{\infty} a_n(x-c)^n$ have radius of convergence $R_1 > 0$ and $\sum_{n=0}^{\infty} b_n(x-c)^n$ have radius of convergence $R_2 > 0$. Then the series

$$\sum_{n=0}^{\infty}(a_n + b_n)(x-c)^n$$

converges and satisfies

$$\sum_{n=0}^{\infty}(a_n + b_n)(x-c)^n = \sum_{n=0}^{\infty} a_n(x-c)^n + \sum_{n=0}^{\infty} b_n(x-c)^n$$

for all x satisfying $|x-c| < \min\{R_1, R_2\}$.

In general, it can happen that the radius of convergence R for the series $\sum_{n=0}^{\infty}(a_n + b_n)(x-c)^n$ is strictly bigger than $\min\{R_1, R_2\}$. The analogous remark applies to the product of two series given in the next subsection.

Example 50 *Consider* $\sum_{n=0}^{\infty} n!x^n$ *and* $\sum_{n=0}^{\infty}(-n!)x^n$. *Thus,* $a_n = n!$, $b_n = (-n!)$. *Then* $R_1 = R_2 = 0$. *However,* $\sum_{n=0}^{\infty}(a_n + b_n)x^n = \sum_{n=0}^{\infty} 0x^n$, *with* $R = \infty$.

4.3.2 Term by term multiplication of power series

Let $\sum_{n=0}^{\infty} a_n(x-c)^n$ have radius of convergence $R_1 > 0$ and $\sum_{n=0}^{\infty} b_n(x-c)^n$ have radius of convergence $R_2 > 0$. Then for all x satisfying $|x-c| < \min\{R_1, R_2\}$, it holds that

$$\left(\sum_{n=0}^{\infty} a_n x^n\right)\left(\sum_{n=0}^{\infty} b_n x^n\right) = \sum_{n=0}^{\infty} c_n x^n,$$

where

$$c_n = a_0 b_n + a_1 b_{n-1} + \cdots + a_{n-1} b_1 + a_n b_0.$$

4.4 Differentiability of power series

Consider a function f that is defined on an interval say $I \subseteq \mathbb{R}$ containing x_0. The function f is said to be differentiable at x_0 if $\lim_{h\to 0} \frac{f(x_0+h)-f(x_0)}{h}$ exists and is finite. In this case, denote $f'(x) = \lim_{h\to 0} \frac{f(x_0+h)-f(x)}{h}$ and call it the derivative of f at x_0. Other notations for $f'(x_0)$ include $\frac{df}{dx}(x_0)$ and $f^{(1)}(x_0)$. The last notation will be often used in this chapter.

At this stage, the above limit has not been defined. There are two equivalent ways of defining it. One is due to Cauchy, and one is due to the German mathematician Heine. Heine's definition involves convergence of sequences, and is thus understandable given the materials covered so far, and we formulate it as follows:

$\lim_{h\to 0}\frac{f(x_0+h)-f(x_0)}{h} = A \in \mathbb{R}$ if and only if $\lim_{n\to\infty}\frac{f(x_0+h_n)-f(x_0)}{h_n} = A$ for each sequence $\{h_n\}$ converging to 0 with $h_n \neq 0$ and $h_n \in I$.

It is assumed that the derivative of common elementary functions are known to you. Usually, they are taught at high school. For example, if $f(x) = ax^n$, then $f'(x) = anx^{n-1}$, $\sin' x = \cos x$, $\cos' x = \sin x$; if $f(x) = e^x$, then $f'(x) = e^x$; if $f(x) = \ln(1+x)$, then $f'(x) = \frac{1}{1+x}$ for $x > -1$. Also, the standard rules of differentiation are assumed to be known to you, such as the chain rule, product rule, etc.

A function f is said to be differentiable on an interval I if it is differentiable at every point in that interval. In this case, f' itself defines a function on I. If f' is differentiable on I, then its derivative is called the second order derivative of f, denoted by $f^{(2)}$ or f''. In general, the derivative of the $n-1$st order derivative of f is called the nth order derivative of f. We denote by $f^{(n)}$ the nth order derivative of f, provided that it exists. We put $f^{(0)} = f$.

Definition 24 (Infinitely many times differentiable function) *If for each $n \in \mathbb{N}$, $f^{(n)}$ exists on I, then f is said to be infinitely many times differentiable on I.*

Theorem 41 (Differentiability of power series) *Suppose $\sum_{n=0}^{\infty} a_n x^n$ is a power series with radius of convergence R. Then, the power series $\sum_{n=1}^{\infty} na_n x^{n-1}$ has radius of R, too. Moreover, the function $f : (-R, R) \to \mathbb{R}$ given by $f(x) = \sum_{n=0}^{\infty} a_n x^n$ is infinitely many times differentiable on $(-R, R)$. In particular, $f'(x) = \sum_{n=1}^{\infty} na_n x^{n-1}$ for all $x \in (-R, R)$.*

4.5 Maclaurin and Taylor Series

Definition 25 (Taylor series) *For a function f, if $f^{(n)}(c)$ is defined for all $n \in \{0, 1, \dots\}$, then the power series*

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!}(x-c)^n = f(c) + \frac{f^{(1)}(c)}{1!}(x-c) + \frac{f^{(2)}(c)}{2!}(x-c)^2 + \frac{f^{(3)}(c)}{3!}(x-c)^3 + \cdots$$

is called the Taylor series of the function f about c.

This terminology is after the English mathematician Taylor.

Theorem 42 *Let $c \in \mathbb{R}$ be fixed. Consider the function f defined on $(c-R, c+R)$ by $f(x) := \sum_{n=0}^{\infty} a_n(x-c)^n$, where $R \in (0, \infty]$ is a possibly infinite constant. Then*

$$a_n = \frac{f^{(n)}(c)}{n!}$$

for each $n \in \mathbb{N} \cup \{0\}$. Consequently, if the values of a function f equal the values of a power series over $(c-R, c+R)$, in which case we say f has the power series representation or expansion about c, then that power series must be the Taylor series of f about c.

Proof. Note that $f^{(0)}(c) = f(c) = a_0$, i.e., $a_0 = \frac{f(c)}{0!}$.

Next, on $(c-R, c+R)$, after applying repeatedly Theorem 41 to differentiating

$$f(x) = \sum_{n=0}^{\infty} a_n(x-c)^n$$

m times, we get

$$f^{(m)}(x) = \sum_{n=m}^{\infty} n(n-1)\cdots(n-m+1)a_n(x-c)^{n-m},$$

so that

$$f^{(m)}(x) = m!a_m + \sum_{n=m+1}^{\infty} n(n-1)\cdots(n-m+1)a_n(x-c)^{n-m}$$

whenever $|x-c| < R$. Setting $x = c$ in this last expression we see that the terms in the sum for $n \geq m+1$ vanish, leaving

$$f^{(m)}(c) = m!a_m;$$

i.e. $a_m = \frac{f^{(m)}(c)}{m!}$, as claimed. □

Consider a function f defined on an interval $I \subseteq \mathbb{R}$. Assume that f is infinitely many times differentiable at on I, so that $f^{(n)}$ is defined on I for all $n \in \mathbb{N}$. Recall also that $f^{(0)} := f$.

Let $c \in I$. Then one can consider the Taylor series of f centered at c. Note that it can happen that $f(x) \neq \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!}(x-c)^n$ on I. E.g., it can happen that for some $x \in I$, $\sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!}(x-c)^n$ does not even converge.

It is of interest to investigate when $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!}(x-c)^n$ at least when x is close enough to c.

This question is beyond the scope of this module. But we point out the general direction of investigations.

It will be seen in a subsequent module that for the function f under consideration with the constant $c \in I$, for each $m \in \mathbb{N}$ and for each $x \in I$,

$$f(x) = \sum_{n=0}^{m} \frac{f^{(n)}(c)}{n!}(x-c)^n + r_n(x),$$

where $r_n(x)$ is called the remainder term. You will see several alternative forms of this remainder term.

Then if we can show $\lim_{n\to\infty} r_n(x) = 0$ for all x that is sufficiently close to c, then we would get $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!}(x-c)^n$, i.e., the function f has a Taylor series expansion about c.

Definition 26 (Maclaurin series) *For a function f, if $f^{(n)}(0)$ is defined for all $n \in \{0, 1, \dots\}$, then the power series*

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n = f(0) + \frac{f'(0)}{1!}x + \frac{f^{(2)}(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \cdots$$

is called the Maclaurin series of the function f.

This terminology is after the Scottish mathematician Maclaurin.

4.6 Taylor and Maclaurin series of common functions

1. For any $x \in \mathbb{R}$

$$e^x = 1 + x + \frac{x^2}{2} + \cdots + \frac{x^n}{n!} + \cdots = \sum_{n=0}^{\infty} x^n/n!.$$

2. For any $x \in \mathbb{R}$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} + \cdots = \sum_{n=0}^{\infty} (-1)^n x^{2n+1}/(2n+1)!.$$

3. For any $x \in \mathbb{R}$

$$\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{4!} + \cdots + (-1)^n \frac{x^{2n}}{(2n)!} + \cdots = \sum_{n=0}^{\infty} (-1)^n x^{2n}/(2n)!.$$

4. For any $\alpha \in \mathbb{R}$ and x such that $|x| < 1$

$$\begin{aligned}(1+x)^\alpha &= 1 + \alpha x + \frac{\alpha(\alpha-1)}{2}x^2 + \cdots + \frac{\alpha(\alpha-1)\dots(\alpha-n+1)}{n!}x^n + \cdots \\ &= \sum_{n\geq 0} \frac{\alpha(\alpha-1)\dots(\alpha-n+1)}{n!}x^n.\end{aligned}$$

5. From 4. we have, for any x such that $|x| < 1$,

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots + x^n + \cdots = \sum_{n=0}^{\infty} x^n;$$
$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \cdots + (-1)^n x^n + \cdots = \sum_{n=0}^{\infty} (-x)^n.$$

6. For any x such that $-1 < x \leq 1$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots + (-1)^{n-1}\frac{x^n}{n} + \cdots = \sum_{n\geq 1} (-1)^{n-1}\frac{x^n}{n}.$$

4.7 Examples

Example 51 *(a) Calculate the Maclaurin series of $f(x) = e^{-3x}$.*

(b) Calculate the Maclaurin series of $f(x) = \cos(2x)$.

Solution. (a) We know that

$$e^t = 1 + t + \frac{t^2}{2} + \cdots + \frac{t^n}{n!} + \cdots$$

Putting $t = -3x$, we get

$$e^{-3x} = 1 - 3x + \frac{9}{2}x^2 - \frac{9}{2}x^3 + \cdots + \frac{(-3)^n}{n!}x^n + \cdots$$

The right hand side is the required Maclaurin series. By the way, the expansion is valid for all x.

(b) We know that

$$\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{4!} + \cdots + (-1)^n \frac{x^{2n}}{(2n)!} + \cdots$$

Replacing x with $2x$, we get

$$\begin{aligned}\cos 2x &= 1 - \frac{(2x)^2}{2} + \frac{(2x)^4}{4!} + \cdots + (-1)^n \frac{(2x)^{2n}}{(2n)!} + \cdots \\ &= 1 - \frac{4x^2}{2} + \frac{16x^4}{4!} + \cdots + (-1)^n \frac{2^{2n}x^{2n}}{(2n)!} + \cdots\end{aligned}$$

The right hand side is the required Maclaurin series. By the way, the expansion is valid for all x.

Example 52 *Calculate the Maclaurin series of $f(x) = e^{-3x} - \cos(2x)$ as far as terms involving x^4.*

Solution. We have

$$\begin{aligned}e^{-3x} &= 1 - 3x + \frac{9}{2}x^2 - \frac{9}{2}x^3 + \frac{(-3)^4}{4!}x^4 + \cdots \\ \cos 2x &= 1 - \frac{4x^2}{2} + \frac{16x^4}{4!} + \cdots\end{aligned}$$

so that

$$\begin{aligned}f(x) &= e^{-3x} - \cos(2x) \\ &= \left(1 - 3x + \frac{9}{2}x^2 - \frac{9}{2}x^3 + \frac{(-3)^4}{4!}x^4 + \cdots\right) \\ &\quad - \left(1 - \frac{4x^2}{2} + \frac{16x^4}{4!} + \cdots\right) \\ &= -3x + \frac{13}{2}x^2 - \frac{9}{2}x^3 + \frac{65}{24}x^4 + \cdots\end{aligned}$$

Example 53 *Calculate the Maclaurin series of* $f(x) = e^{-3x}\cos(2x)$ *as far as terms involving* x^4.

Solution. It is often easier to work directly, rather than use the above (Cauchy product) formula. We know

$$\begin{aligned} e^{-3x} &= 1 - 3x + \frac{9}{2}x^2 - \frac{9}{2}x^3 + \frac{(-3)^4}{24}x^4 + \cdots \\ \cos 2x &= 1 - \frac{4x^2}{2} + \frac{16x^4}{24} + \cdots \end{aligned}$$

so that

$$\begin{aligned} f(x) &= e^{-3x}\cos(2x) \\ &= \left(1 - 3x + \frac{9}{2}x^2 - \frac{9}{2}x^3 + \frac{(-3)^4}{24}x^4 + \cdots\right)\left(1 - \frac{4x^2}{2} + \frac{16x^4}{24} + \cdots\right) \\ &= 1 - 3x + \left(\frac{9}{2} - \frac{4}{2}\right)x^2 + \left(\frac{12}{2} - \frac{9}{2}\right)x^3 + \left(-\frac{36}{4} + \frac{16}{24} + \frac{(-3)^4}{24}\right)x^4 + \cdots \\ &= 1 - 3x + \frac{5}{2}x^2 + \frac{3}{2}x^3 - \frac{119}{24}x^4 + \cdots \end{aligned}$$

Example 54 *(a) Find the Taylor series for* $f(x) = e^x$ *about* 1.

(b) Calculate the Taylor series of $f(x) = \dfrac{1}{3x+2}$ *about* -2.

Solution. (a) We know that

$$e^t = 1 + t + \frac{t^2}{2} + \cdots + \frac{t^n}{n!} + \cdots$$

for any t. We want an expansion in terms of $(x-1)$. Putting $t = (x-1)$ in the above, we get

$$\begin{aligned} e^x = e^{(x-1)+1} = ee^{(x-1)} &= e\left(1 + (x-1) + \frac{(x-1)^2}{2} + \cdots + \frac{(x-1)^n}{n!} + \cdots\right) \\ &= e + e(x-1) + \cdots + \frac{e}{n!}(x-1)^n + \cdots \end{aligned}$$

(b) We know that for any x such that $|x| < 1$,

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots + x^n + \cdots = \sum_{n=0}^{\infty} x^n$$

Now we want an expansion involving powers of $(x - -2) = (x+2)$. Since

$$3x + 2 = 3(x+2) - 6 + 2 = 3(x+2) - 4 = -4\big(1 - 3(x+2)/4\big)$$

we have, replacing x by $3(x+2)/4$ in the expansion of $1/(1-x)$

$$\begin{aligned} \frac{1}{3x+2} &= -\frac{1}{4}\frac{1}{1 - 3(x+2)/4} \\ &= -\frac{1}{4}\left(1 + \frac{3(x+2)}{4} + \frac{9(x+2)^2}{16} + \cdots + \frac{3^n(x+2)^n}{4^n} + \cdots\right) \\ &= -\frac{1}{4} - \frac{3}{16}(x+2) - \cdots - \frac{3^n}{4^{n+1}}(x+2)^n + \cdots \end{aligned}$$

The expansion is valid for $|3(x+2)/4| < 1$, i.e. for $|x+2| < 4/3$.